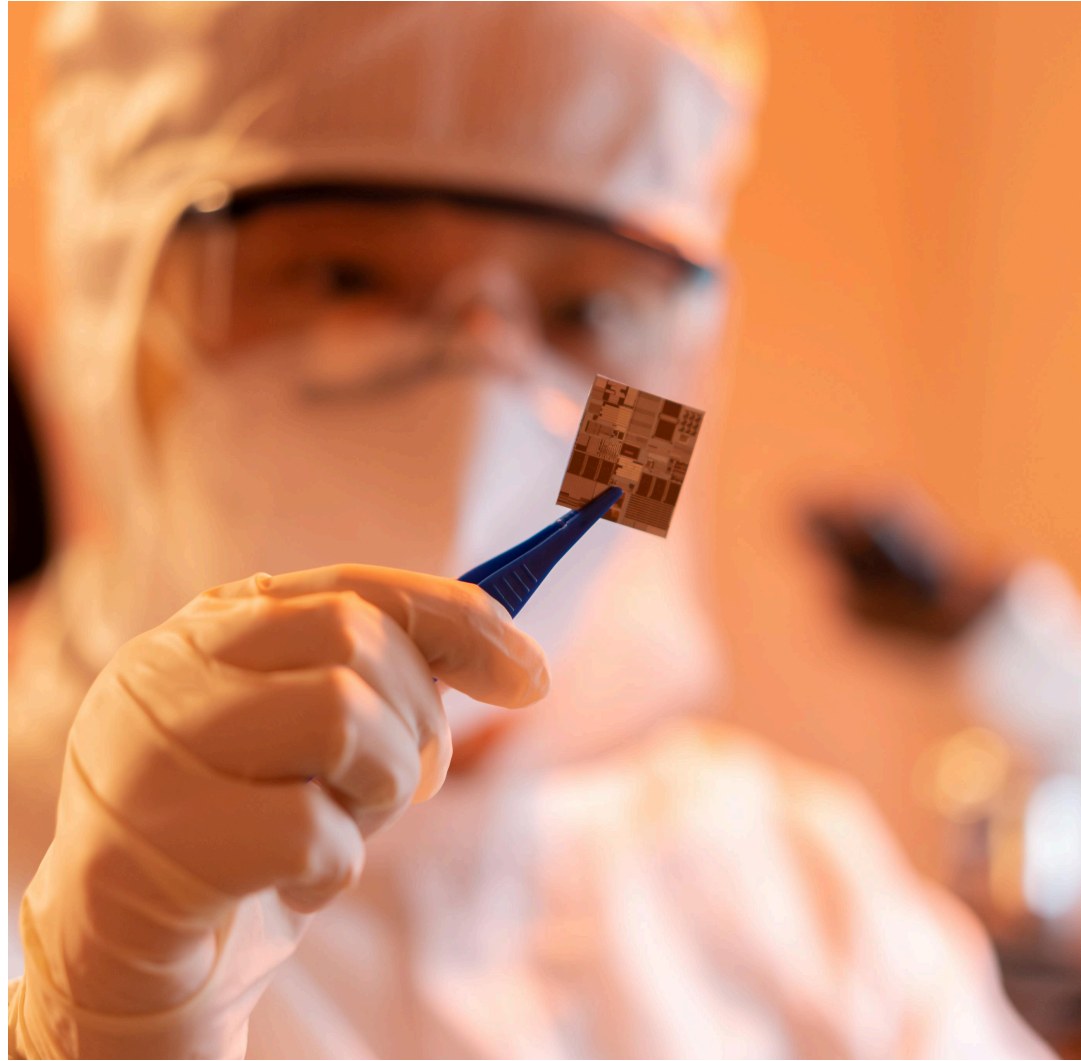


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LAURA RYAN
ALVIN DJAJADIKERTA
JAMES PHILLIPS
BENEDICT MACON-COONEY
JAKOB MÖKANDER



A New National Purpose: Reimagining UK Science and Technology through Lovelace Disruptive Invention Laboratories

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Foreword

As Nobel laureate Paul Romer put it, “Perhaps the most important ideas of all are meta-ideas – ideas about how to support the production and transmission of other ideas ... Only a failure of imagination – the same failure that leads the man on the street to suppose that everything has already been invented – leads us to believe that all of the relevant institutions have been designed and all of the policy levers have been found.” The United Kingdom now has an opportunity to act on a meta-idea of its own: build new types of institutions that enable disruptive discovery and invention at scale.

Disruptive research is the upstream source of long-run growth: it opens new scientific frontiers, seeds tomorrow’s industries and strengthens national resilience. If the UK wants more of it, the UK must design for it. One of my favourite British sayings is “Horses for courses.” The UK’s universities are among the best in the world and must remain so, but no single model for conducting research is suited to all types of scientific project. The country also needs experimentation and diversity in how it supports research and invention.

This is why I am a strong supporter of the Tony Blair Institute for Global Change’s proposal for the UK to invest in Lovelace disruptive invention labs. The premise is simple: concentrate ambitious scientists and engineers around a shared vision; free them from short-term grant cycles; give them the tools, technical talent and authority to iterate quickly; and judge them on progress towards field-building rather than publication. Do that and you change what’s possible.

I have seen first-hand the value of increasing the structural diversity of the research ecosystem. In 2019 – while serving as the chief innovation officer at Schmidt Futures – I got an email from Sam Rodriques, then a graduate student, including a chapter from his PhD thesis. In it he called for the creation of a new kind of research entity: focused research organisations (FROs). FROs are designed to tackle defined, time-limited scientific projects

that need significant funding, a focused approach and interdisciplinary collaboration, but are not suited to being supported by academic or for-profit mechanisms.

One of Sam's collaborators on the idea, Adam Marblestone, told me that this idea was sufficiently important that he would leave DeepMind and devote the next chapter of his career to making it happen. Adam and I interviewed more than 100 scientists and engineers to identify bottlenecks that could be addressed with an FRO. With generous support from Eric Schmidt and more than 30 other philanthropists, Adam and his co-founder Anastasia Gamick have launched Convergent Research, and ten FROs. These "non-profit startups for science" are pursuing ambitious goals, such as reducing the cost of mapping the brain by a factor of 100, tapping the genetic diversity of the biosphere and powering a revolution in formal mathematics. The UK government's Department for Science, Innovation & Technology, meanwhile, has funded a UK-based FRO called Bind Research.

Most exciting is how the existence of this new model has raised the ambition of the research community. Prior to the creation of FROs, researchers were not asking themselves, "What scientific breakthroughs could I enable as the CEO of an industrial-grade non-profit, with a team of 20 to 30 scientists and engineers, all laser-focused on a single goal?" Now I attend conferences where a researcher I have never met will ask, "Will solving this problem require an FRO?"

By launching Lovelace disruptive invention labs, the UK can pioneer a new research model and raise the ambition of the whole system. Then a new question could be asked: "What scientific horizons can be explored in vision-oriented labs that integrate discovery with engineering, support risk and long-term projects, and give researchers the autonomy and tools to lead?" In answering that question, the UK can set the pace for how countries turn scientific breakthroughs into transformative new fields and industries, strengthening its position as a scientific superpower and delivering prosperity.

Tom Kalil *CEO of Renaissance Philanthropy and chair of Convergent Research; former deputy director of policy in the Office of Science and Technology Policy under President Barack Obama*

Executive Summary

At moments of rapid technological change, scientific and technological leadership are not simply levers of prosperity – they are foundations of sovereignty. Countries that lead on creating new capabilities and scientific fields gain both economic rewards and the ability to set the terms of global progress. If the United Kingdom is to influence how the inventions of tomorrow are developed, deployed and governed, it must be home to the laboratories where that future is founded. That means making the UK the best place in the world to pursue frontier science and build transformative technologies into new industries.

History shows what is possible. Revolutions at the intersection of science and technology, such as those of telecommunications, molecular biology and personal computing, were some of the most transformative forces of the 20th century. Each meant profound shifts in humanity's understanding of nature and the ability to harness its powers for progress. The features and origins of such revolutions have been the subject of considerable study, as scholars seek to understand how the type of research that sparked them can be intentionally fostered to enhance national prosperity, health and security. From this body of work, a consistent insight emerges: that disruptive leaps are often the product of distinctive institutional conditions – and that such conditions can, in principle, be designed.

The next wave of revolutions is already underway; the question is whether the UK will shape them or be shaped by them.

The Tony Blair Institute for Global Change's [New National Purpose](#) series has laid the policy foundations for a reimagined state – one that places science and technology at the heart of national renewal. Previous reports have already informed government policy, including the creation of an internationally open AI security institute, the appointment of domain experts to ministerial roles and ten-year R&D spending settlements.

This report develops a core recommendation of those previous reports: that the UK should create a new kind of research laboratory at scale – built to support agency, risk and the exploration of long-term research visions at the intersection of science and technology. We term these Lovelace disruptive invention labs. This is a critical step if the country is to lead on creating the industries of tomorrow, as well as reform the culture of early-stage science and technology research to make it more entrepreneurial and dynamic, filling a global niche for new career paths for the most sought-after talent.

In creating purpose-built environments for visionary technoscience, the UK would pioneer a new kind of “organisational technology” akin to the invention of the corporation, the university and the venture capital firm – or more recently, the focused research organisation (FRO). These institutional innovations unlocked new ways to coordinate talent, mobilise capital and solve scaling challenges, enabling step changes in capability.

The Opportunity

Scientific progress takes many forms. Some is linear and cumulative: solving well-defined problems, refining established models and scaling proven techniques. But some is disruptive: it introduces entirely new conceptual frameworks and exists not within or across existing fields and paradigms, but rather establishes new ones. These different modes of discovery and invention generally thrive under different conditions, demanding different kinds of institutional support. What works well for advancing mature fields may not be suited to deep exploration of the unknown. A system optimised for peer review and steady publication may not support long-horizon speculative research, high-risk projects and bold work that defies prevailing consensus.

The UK’s public-research ecosystem is world-class in many respects, but it is also unusually concentrated around one type of institution: the university lab. Compared to other countries that lead in science and technology, the UK relies heavily on academia not only for basic research, but also for invention, translation and early-stage technology development.

This model has produced global excellence, but it is not designed to do everything. In general, it is particularly ill suited to the kind of work that has historically driven the emergence of new industries: high-risk, long-term interdisciplinary research that blends science and engineering, fundamental inquiry and invention, theory and tool-building – research that is not solely discovery science, but also not closely linked to commercial viability. It is also a model that is increasingly failing a generation of talent. Too many of the brightest scientists are walking away, frustrated by bloated bureaucracy and disillusioned by the poor incentives created by a “publish or perish” culture – or deprived of meaningful autonomy until late in their careers.

Shifts in research and immigration policy in the United States have opened a rare window for the UK to position itself not just as an enabler of native talent, but also as a global magnet for scientific brilliance. But that will depend on the UK’s ability to offer something fundamentally different: labs built to support agency, long-term stability and exploration of the scientific frontier. The country needs a new kind of public-research institution, purpose built for a synergy of transformative theory, discovery and invention.

A disproportionate amount of disruptive, field-creating work has emerged from a handful of unconventional scientific institutes, including Building 20 at the Massachusetts Institute of Technology, the Laboratory of Molecular Biology (LMB), Bell Labs, the Los Alamos National Laboratory and Xerox PARC. More recently, a similar phenomenon has been emerging with the relatively concentrated progress in the development of early-stage artificial intelligence. Much variety exists between these organisations, in both form and purpose. But many share a few core principles and internal structures uniquely suited to disruptive work, often differing markedly from those that dominate in UK publicly funded research.

- A focus on vision-oriented research, organised around ambitious and broad scientific missions rather than narrow objectives, historical disciplines or specific commercialisable goals.
- The integration of scientists and engineers under one roof, with equal emphasis on the development of tools and prototypes alongside scientific discovery.

- Core funding to provide protection from external pressures, such as the need to publish and secure consensus-driven grants on short timescales.
- Smaller teams and flattened hierarchies, resulting in the empowerment of junior researchers and a culture encouraging agency and risk-taking.
- Long-term stable financial support at an institutional level, removed from immediate commercial pressures.

While elements of these exist within the UK public R&D ecosystem, nowhere are all – or even most – of these features found together. Where they are found in existing institutes, the labs in question are often subject to the high-level homogenising influences of the UK's university-oriented landscape, inheriting its incentives, career structures and bureaucratic audit culture. The UK should pioneer a national network of Lovelace disruptive invention labs purpose-built to address these limitations and seize a global opportunity.

SPOTLIGHT

Ada Lovelace and Her Scientific Vision

Ada Lovelace was a 19th-century mathematician who is widely acknowledged as the world's first computer programmer. This is due to her work with inventor and engineer Charles Babbage, specifically on the concept of the analytical engine (the precursor to modern computers), for which she wrote notes on codes and instructions.

Lovelace dreamt of a “poetical science” for a better world – a creative venture at odds with the modern habit of treating science as sterile procedure. She believed that symbols could be taught to sing, and that the analytical engine might move beyond barren computation to “act upon other things besides numbers ... the engine might compose elaborate and scientific pieces of music of any degree of complexity or extent”. She foresaw by a century the computer revolution and today's technological society, sketching possibilities including humans taking to the air, influencing the mind-body relationship and enhancing thought itself. It is easy to forget that the planes in the sky and the medicines in human cells were once nothing more than visions in the minds of the likes of Lovelace, so the Lovelaces of today are too seldom cultivated.

There is a key need for junior researchers globally to pursue long-term research programmes, self-organise and build new fields into industries, as they could in the postwar period that created our technological world. To do this, we must reinvent discovery, using Lovelace's creative ethos to pioneer new and complementary ways to conduct the art of research. We use her

name not as ornament, but as an injunction to rekindle the spirit that sparked the creation of the Royal Society more than three centuries ago – and in doing so, make the UK the home of today’s poetical science.

The Vision

The proposed network of Lovelace disruptive invention labs should sit outside of – but be complementary to – academia, taking its design cues from the striking commonalities between historically transformative labs – as well as more recent institutional successes such as DeepMind and the Arc Institute – while responding to the contemporary structural constraints of the UK’s public R&D landscape.

Lovelace labs would provide intentionally and distinctly different environments, incentives and career structures to those of academia. Each lab would be organised around a bold, unifying research vision – not a discipline or narrow goal, but a new horizon of possibility. They would be physically co-located, cross-disciplinary communities, where theory, discovery and engineering happen under one roof. Long-term core institutional funding would support autonomy and exploration, while each lab would be led by an empowered director, and internal review would replace grant-chasing and short-term audits.

The minimal elements above are, in our view, necessary foundations. Beyond them, the precise configuration of a Lovelace lab should be up to its director and should be field-specific, evolving over time. The guiding principle should be to learn by doing. Using this approach, a new public-research entity – the Lovelace Society – would oversee the Lovelace lab network. It would act as a supportive platform (akin to the Max Planck Society) by providing a set of

freedoms, guardrails and shared principles, with deliberate flexibility for labs to adapt. Lab directors would be hired for their research vision, organisational judgement and commitment to a new way of doing science and technology research; the centre would supply the support structure and oversight.

What follows is one possible configuration that could guide the first generation of Lovelace labs. We set it out not as a definitive template, but as a starting point for experimentation.

We propose that early labs should be centred around a Lovelace Fellowship, which would offer long-term renewable support to promising junior researchers so that they could pursue bold ideas from day one. Rather than emulating the academic career pathway – in which creative freedom is too often stymied by slow career progression, hierarchy, the pursuit of funding and managerial burdens – Lovelace fellows would not apply for grants and would be free of the administrative constraints of conventional postdoctoral or principal investigator (PI) roles. These fellowships would invert the traditional model by providing substantial resources and independence much earlier in a career, and encouraging hands-on, team-based invention rather than top-down management.

To ensure that small, self-organising groups could scale efforts when necessary, labs would also support medium-sized coordinated research programmes: structured initiatives with dedicated budgets, more defined objectives, and targeted coordination to tackle problems that require sustained collective effort. These would be akin to scaled-down, in-house versions of FROs. Research outputs would be shared via open, AI-native platforms, and labs would host workshops and training programmes (in collaboration with industry) in “edge-of-the-art” knowledge and techniques, to help build new fields from the ground up.

With the right support, Lovelace labs could become global magnets for talent, engines of discovery and a powerful new tool in the UK’s long-term strategy to lead the industries and scientific revolutions of the future.

A New Role for Government

Realising this vision would require an actor with the scale, patience and risk appetite to create such institutions – and in the UK, this role would likely need to be played by the government. The landscape that once supported disruptive research has fundamentally shifted. Many pioneering 20th-century labs operated as the research arms of companies with huge monopolies, which could afford to invest in exploratory science over decadal time-horizons, without immediate commercial pressures constraining agendas. In today's highly competitive markets, few private entities can afford this kind of investment, and the model survives only among a handful of technology giants.¹

In the UK, government is best placed to take up this mantle. It is in a unique position to provide patient, long-term capital combined with a tolerance for risk. It can also adopt an unusually broad approach to value capture – recognising the substantial economic and social benefits that emerge from cultivating innovation clusters, regional agglomeration and new industrial ecosystems. This is particularly true outside the US, where countries lack domestic tech giants.

Recommendations

The UK should establish a national network of Lovelace disruptive invention labs. Collectively they should be tasked with the pursuit of long-term missions at the intersection of science and technology that have the potential to seed new industries for the UK. Initially, the goal should be to create two to three major labs with globally competitive resourcing, alongside a set of smaller affiliate labs. These labs would be overseen by a new public body – the Lovelace Society – with a mandate to create, sustain and coordinate the network. To succeed it must operate with substantial autonomy, secure long-term funding and be underpinned by the right structural foundations.

1. **Legislate to create the Lovelace Society as a dedicated public entity.**

Existing structures such as UK Research and Innovation (UKRI) and the Advanced Research and Invention Agency (ARIA) are not well suited to the distinct mission of the network of Lovelace labs, which requires a dedicated structure with its own mandate, tailored specifically to the aim of building long-term research communities with a high degree of autonomy and the ability to manage complex infrastructure. A bespoke entity ensures the necessary operational freedom and mission alignment. It also allows for specific legal exemptions from standard governmental operating procedures (such as procurement rules), replacing them with purpose-built systems. We recommend that the UK government legislate for the creation of a new research entity – the Lovelace Society – and fund it sufficiently for 15 years of operations. The legislation should be closely modelled on the 2022 Advanced Research and Invention Agency Act, with the Lovelace Society reporting to the Department for Science, Innovation & Technology in a similar manner to ARIA. By 2040, a relatively small percentage of the annual public R&D budget, in the order of 3 to 4 per cent, should be invested in the Lovelace labs network. This should support the establishment and maintenance of several major global labs and smaller affiliated initiatives.

2. **Create an initial founding unit to launch the Lovelace Society.** Before the society can oversee its first labs, a small, time-limited founding unit should be set up to: scope potential research areas (nascent windows of technoscientific opportunity) and lab directors; identify and recruit the society's leadership; assemble the society's expert board; and refine the Lovelace Society's broad operating model and essential foundational principles and freedoms, leaving substantial latitude for evolution, and for lab directors to pursue their own visions. The UK government should allocate a £3 million initial grant for a three-year founding unit.

3. **Secure sustained funding.** To attract top talent, support high-risk work and move beyond the setup phase with confidence, labs would need support that is committed for at least 15 years. A single business case for the Lovelace Society should be approved that outlines the mission that the funding has been set aside to achieve, with the society's board and subsequent external reviews overseeing more granular spend. About

£250 million per year would be needed to set up at least two to three labs – funded to compete at the global frontier, akin to the Arc Institute – alongside a group of smaller, affiliate labs. This represents about 1.25 per cent of the UK R&D budget in 2025 terms – an initial commitment that would rise over time as the network scales towards a 3 to 4 per cent end-state by 2040. This £250 million per year estimate is conservative, as it does not factor in the additive benefit of crowding in joint investment with philanthropic and for-profit capital, as occurred with the previous government’s Research Ventures Catalyst. We emphasise that it is better to fund a smaller number of well-funded labs than to set up a greater number with insufficient resources.

4. **Legislate for the integration of existing labs.** Some high-potential scientific programmes (from universities or other government research, for example) outgrow their institutional homes, but lack the legal, financial and operational structures to scale and become mission-aligned physical institutes. It should be easy to transfer legal and funding control of projects and labs from the existing infrastructure to the Lovelace Society. The UK government should commission legal and policy work to investigate what legal carve-outs would be necessary for this integration.

This report differs substantially from previous New National Purpose papers in style, scope and target audience. It is a combination of historical analysis, policy recommendation, implementation plan and manifesto, which together provide a path to create a new kind of research structure in the UK over a 15-year time horizon.

A Vision for UK Science and Technology

The United Kingdom is at a critical juncture in its approach to science and technology, and to the role of R&D in the economy. At this moment in history, new platform technologies such as artificial intelligence are unlocking potential new industries. Strong first-mover advantages in technology mean that those countries that pioneer these industries and use them to drive prosperity will be the ones that flourish in the 21st century. To use businessman and science philanthropist David Sainsbury's phrase, "windows of opportunity" have opened.²

Rising to this moment requires both greater ambition and a renewed approach. Previous [New National Purpose](#) reports from the Tony Blair Institute for Global Change, as well as others from the likes of Onward and UK government reviews, have set out critical short-term challenges and recommendations for incremental improvements and reforms.^{3,4,5,6} But the UK also needs a bold ten-year vision to regain true leadership status in the world of science and technology. This aspect of national strategy is the focus of this report. Without such an agenda, the country risks continuing to lurch from one short-term stopgap to another – its approach dominated by reactive tactics without a cohesive strategy, and its portfolio responding to short-term stakeholder pressures.

In this paper we advocate for the UK investing a small percentage of its public R&D budget (in the order of 3 to 4 per cent by 2040) in pioneering a complementary approach to research at the intersection of science and engineering (technoscience), through the creation of a network of physical laboratories. These should draw inspiration from the most impactful institutes of the past and from those pioneering AI today. Their aim would be to reestablish the UK as the place where transformative new industries are born, attracting brilliant and diverse junior talent from around the world, including those disillusioned or disenfranchised by traditional approaches.

The recommendations in this report should be pursued alongside an elevated public R&D budget, as well as broader reform. TBI does not claim that the precise form of this proposal is the perfect or permanent solution; rather, we hope that others will continue to improve upon it to reinvent the country's pursuit of discovery.

Challenges and Opportunities in UK R&D

Since the mid-20th century, publicly funded scientific research has been a principal instrument of progress. Yet after half a century of expanding inputs, scientific breakthroughs are occurring with diminishing efficiency. Much of the system's output has become incremental, with progress arriving in smaller steps for greater effort.

As outlined by Patrick Collison and Michael Nielsen in *The Atlantic*, the scientific enterprise has grown dramatically since the 20th century.⁷ Significantly more money and time is spent on research, and there is a far greater number of researchers – working in ever larger lab groups – who publish more papers.^{8,9} Yet the payoff has not kept pace: numerous studies echo the fact that returns on investment in traditional approaches have diminished over time.^{10,11} Productivity has declined across fields and industries,^{12,13} funding has become more conservative¹⁴ and both research papers and patents have become less disruptive – according to analysis in *Nature* they are “increasingly less likely to break with the past in ways that push science and technology in new directions”.¹⁵ Large-scale evidence suggests that as fields expand, citation dynamics entrench existing orthodoxies, while novel work can struggle to gain support or attention.¹⁶

Public research in the UK is no exception to these headwinds. The UK government has substantially increased R&D funding over the past decade, with commitments to raise economy-wide R&D spend to 2.4 per cent of GDP and public R&D to £22 billion a year by 2027 (though some of this is eroded by inflation and the post-2020 absorption of Horizon Europe costs).¹⁷ The rationale is clear. Innovation has historically done heavy lifting for UK productivity: estimates used in parliamentary evidence attribute about 51 per

cent of labour-productivity growth between 2000 and 2008 to it.^{18,19} Reviews conclude that public R&D delivers high social returns, and also find higher GDP multipliers for public R&D than generic public spending;²⁰ there is also evidence of significant crowding-in of private contributions.^{21,22} These figures support the expectation that higher R&D spending will translate into faster growth, stronger productivity, geostrategic advantage and the birth of new industries.

Despite this expectation – and the increased sums invested – there are signs that the UK is not yet seeing commensurate improvements in outcomes. Productivity has stagnated,²³ the rate of patent creation has plateaued²⁴ and the birth of new industries or major innovative companies has been limited.²⁵ These are imperfect proxies for research performance, not least because there are lags between increases in public R&D spend and measurable outputs,²⁶ and structural blockers to tech transfer and scaling that may inhibit the creation of new industrial ecosystems. But taken together, they are economy-level signs that warrant consideration.

If higher spending were translating into stronger frontier research performance, this would likely be most evident in recent research outputs. The conventional wisdom that the UK is world-leading in discovery science has merit: it has a first-class public-research base and generates impactful and influential scientific research for a country of its size and given its still relatively low investment levels compared to other advanced economies.²⁷ The country ranks first globally for research quality (by field-weighted citation impact) and third in total publications and citations, despite having less than 1 per cent of the world's population. But these measures benchmark publication impact, not frontier capability or disruptive potential. They are the wrong yardstick for a policy goal of assessing and supporting industry-creating, economically transformative research.

On this score, reviews by leading scientists have warned against complacency, noting the fragility of the UK's position at the true frontier of science and technology.^{28,29} In particular, it may be weak in the type of

research and invention that has historically seeded new industries: technology-intensive areas at the intersection of science and engineering.^{30,31}

Recent analysis³² looked at the three priority technology families in the UK's 2021 Integrated Review of national strategy (AI, engineering biology and quantum) and evaluated how the country has been performing at the very top level of research in those fields over the past decade.

- **UK performance is much weaker than assumed when using the most selective measures.** While the UK is responsible for 13 per cent of the top 1 per cent most cited works across fields, this is a relatively unselective measure given the very large volume of research published globally. Looking specifically at the top 100 papers in each priority field (a proxy for the true cutting edge), the UK can rarely lay claim to more than 3 to 6 per cent of them.
- **The UK's most outstanding performers are atypical laboratories.** UK performance at the frontier is strongly driven by a small number of atypical organisations, such as the Cambridge Laboratory for Molecular Biology (LMB) and Google DeepMind. Removing DeepMind, for example, reduces the UK's share of the citations amongst the top 100 recent AI papers from 7.84 per cent to 1.86 per cent. That frontier performance is largely driven by a few 'superstars' may not be surprising.³³ But critically, these outlier labs share distinctive features that differ substantially from the environments that almost all UK public R&D money funds.
- **Individual institutions outperform the entirety of UK academia.** In each case examined, a single institution (such as MIT or DeepMind) matched or exceeded the entire UK academic performance.

Such indicators offer only a partial view, not least because they deal with a low number of examples. Still, they suggest that the UK might not be leading at the most selective frontiers as confidently as assumed, and that it should take deliberate steps to strengthen its position. Further, recent analysis

shows that the apparent upturn in the UK's performance is being driven by increased co-authorship of international papers, which does not necessarily indicate improved domestic performance.^{34,35}

In recent decades the UK, along with other leading science nations, has largely pulled one lever: funnel more money through the existing machinery. Then, when the anticipated results have not materialised, the same lever has been pulled again.

Meanwhile, others – largely philanthropists and tech giants – have pursued a different approach: building institutions with innovative and bespoke models that combine science with engineering, empower talent and move at pace outside traditional academic settings. Examples include DeepMind and the Arc Institute, which are two of the most dynamic and successful environments globally when it comes to translating AI into scientific capability.

These are key benchmarks for the UK's frontier ambitions. The country may have grown comfortable comparing itself to European systems, ignoring a substantial gap between the latter and progress in the world's most successful regions, such as Boston and Silicon Valley. The successes of organisations such as DeepMind and Arc, grounded in part in structural innovation, present an argument for reform. This should challenge an implicit assumption of UK research policy: that routing more money through existing channels is the best way to deliver frontier advances and industry-birthing work.

From Reform to Diversification

There have been some government-initiated efforts to consider how to strengthen the UK's position. Sir Paul Nurse's recent Landscape Review of the UK's research institutions, commissioned by the previous government, highlighted substantial limitations in the current approach to science and technology,³⁶ including:

- **An unbalanced research landscape, globally uniquely dependent on universities.** Eighty per cent of non-business R&D investment in the UK is funnelled into universities. For comparable countries this figure is often between 45 and 60 per cent, indicating that a greater proportion of their investment is in entities like national labs and non-profit organisations. Since the 1980s, R&D expenditure in higher education has gradually increased, while investment in other types of institutions has fallen.
- **Increasing pressure on the academic career path, with particular concerns for early-career researchers.** The default academic career path is under strain due to the number of trainees exceeding available positions by much more than an order of magnitude,³⁷ creating an ever more sharply pyramidal structure (compared by *The Economist* and others to a Ponzi scheme).³⁸ Relatedly, the age of research independence has risen, while the concentration of resources held by individual professors has increased.^{39,40,41}
- **Growth of an overly bureaucratic audit culture within the research system.** Rather than assessing the quality of research over time, too much focus is placed on frequent, burdensome reporting. These reviews often prioritise easily measurable, short-term administrative metrics over the substance or long-term significance of research outputs.

On top of these concerns, we would add a systemic aversion to risk and long-termism driven by homogenous and perverse incentives,⁴² and an absence of vision or mission-oriented research laboratories that place emphasis on integrating discovery and invention under one roof. Relatedly, there is a lack of a role for invention and methods development research at the pre-commercial stage. Nobel laureate and champion of the LMB Sydney Brenner said, “Science proceeds through new methods, new discoveries and new ideas, usually in that order,” yet the pursuit of new methods worldwide is underfunded.

The situation is deteriorating: even prominent principal investigators are struggling to recruit postdoctoral researchers, while calls for structural reform are rising.^{43,44,45} We hear anecdotally, but consistently, that many

and perhaps most of the most talented junior scientists are leaving academia, and often leaving research entirely. The UK's heavy reliance on this model may represent a strategic vulnerability.

These features and trends have not gone unnoticed. They surface in the government's 2021 UK national Innovation Strategy, the Willetts review and the Tickell review, as well as in TBI's [New National Purpose](#) series.

To address these and other challenges, Nurse called for a "revolution by evolution" in the UK's approach: endorsing the need for radical change, but leaving the nature of that change unstated. Core questions remain unanswered. Is the required evolution a gradual process of incremental change, or is it more like the Cambrian explosion,⁴⁶ with the sudden emergence of diverse forms of life? Is it directed evolution, as used in modern synthetic biology⁴⁷ – engineered towards a known goal? And if so, what is that direction? Or is it the "blind watchmaker"⁴⁸ of natural evolution?

In the language of mathematics, evolutionary theory and machine learning, the UK R&D landscape might be stuck on a local optimum while far more effective systems remain out of reach, inaccessible through incremental adjustments that are typically driven by stakeholder pressures. This report is grounded in the view that reform alone will not suffice to meaningfully shift the trajectory of the UK's research system. Many of the challenges are not surface-level inefficiencies but deep structural features, rooted in the core architecture of the R&D system. Rather than refinement, the UK needs diversification: the deliberate cultivation of entirely new forms of the living organisms that are the country's research communities.

The Nurse review, while providing a comprehensive map of the UK's R&D system, stops short of offering the bold recommendations that the moment demands. As noted by Greg Clark, chair of the Commons select committee for science and technology, and echoed by commentary in the *Financial Times*, questions remain as to whether the review and broader government thinking are sufficiently strategic or disruptive to reverse the noted national decline in frontier science and technology.^{49,50} This points to a critical

juncture for UK research policy. Is the country's institutional landscape, with tweaks, fit for purpose? Or does it need substantial new additions and reforms to restore the UK to global leadership?

The UK R&D system has been more or less the same for half a century, with the only major change being a reduction in structural diversity during the Thatcher era, when national labs were closed during budget cutbacks. Today, UK research funding is more likely to be spent within universities than in any other peer country, except Canada.⁵¹ In comparison, the US has well-funded national labs and tech giants that pursue some of the world's most cutting-edge innovation free from normal academic constraints. Germany too has a series of relatively self-governing elite scientific societies. Of particular relevance is the Max Planck Society: an elite basic-science society providing long-term career support and operating parallel to academia, with core federal funding of about €2 billion that is supplemented by generous external income. While this is an imperfect model – and not one to copy wholesale – it demonstrates that durable, globally competitive, government-supported research systems are possible outside of universities.

Given the diverse nature of research as a pursuit, and the unlikelihood that a single model is optimal for all types, we contend that the UK has made a strategic error in relying almost exclusively on one.⁵² That is why we propose prioritising the creation of parallel scientific structures.

Against this backdrop we have set out a proposal grounded in historical analysis, as taking such a view does away with the illusion that the present model is the only one available, or somehow an inevitability. We pair this with a set of policy recommendations and a high-level implementation plan, recognising that several core requirements of our vision run counter to typical government processes.

This report also serves as an introduction to applied metascience for frontline researchers, who may sense that a different way is possible, but are not yet immersed in the emerging possibility space.

Our focus is the creation of a new research entity to complement existing structures. We do not primarily seek to remodel academia; rather we seek to carve out a small space purpose-built for disruptive invention, with the goal of seeding new fields and industries. However, we hope that by highlighting different modes of organising research, broader reform will be encouraged.

Each of the following chapters in this report is relatively self-contained and can be read independently.

- Chapter 2 outlines disruptive science and technology, and the role that this type of research plays in creating new fields and industries.
- Chapter 3 examines landmark laboratories of the 20th century alongside leading private and philanthropic labs today, analysing the features that made them well suited to fostering disruptive technoscience.
- Chapter 4 builds on these insights to set out our vision for a new public-research entity: the Lovelace Society.
- Chapter 5 provides an implementation plan to make this vision a reality.

02

Disruptive Technoscience and the Creation of New Frontiers of Progress

To explore how the UK could build environments oriented towards industry-creating work, a terminology and perspective must be established that can outline the type of research that opens new frontiers. This chapter provides that vantage point: situating transformative work within wider scientific progress, examining its role in technoscientific revolutions and considering how it is supported by institutions. A degree of simplification is inevitable; we have used it to make the essential dynamics legible, rather than to flatten complexity.

While much human progress in science and technology occurs incrementally – through the steady accumulation of knowledge within established disciplines – there are moments when discovery and invention converge to redefine horizons. In this report we use the broad term “technoscience” to mean that tightly coupled mode of progress in which theoretical insight, enabling methods and instruments, and engineered technological systems advance together in short, mutually reinforcing loops. When this iterative process moves quickly within organisations that co-locate discovery and engineering, they can ignite revolutions at the intersection of science and technology.

These technoscientific revolutions – such as those of telecommunications, molecular biology and personal computing – do not simply extend existing trajectories: they open wholly new domains of research, industry and capability. More recently, tight feedback between algorithmic advances, hardware optimisation and practical deployment has propelled AI’s rapid progress in a similar way.

Such revolutions, and the breakthroughs that fuel them, are cornerstones of national and regional economic transformation. For instance, the invention of the transistor at Bell Labs, which emerged from a combination of theoretical

physics, solid-state-materials research and engineering, not only revolutionised telecommunications and electronics but also triggered a cascade of advances that were key to the rise of Silicon Valley as a global technology hub.^{53,54} As of February 2024, companies based in the Silicon Valley area had a collective total-market capitalisation of \$14.3 trillion.

The UK, too, has witnessed how disruptive technoscientific work can seed enduring industrial ecosystems. The Cambridge biotech cluster – now home to more than 600 firms generating more than £7.4 billion in annual revenue and employing nearly 23,000 people⁵⁵ – was critical in nucleating the wider UK biotech sector, laying the groundwork for a thriving industry now worth an estimated £21.3 billion per year.⁵⁶ The origins of this regional success lie in a series of related, foundational and field-generating breakthroughs, from advances in techniques such as X-ray crystallography to the elucidation of the structure of DNA by James Watson and Francis Crick at the Cavendish Laboratory in 1953, which placed Cambridge at the epicentre of the molecular-biology revolution. This discovery and others (for instance, Max Perutz and John Kendrew’s work on protein structures) laid the foundation for the LMB in 1962, where subsequent pioneering research seeded companies such as Cambridge Antibody Technology in 1989, igniting the biotechnology boom that continues to this day.

National governments have a strong interest in making their countries the home of such revolutions. While the direct financial returns on investments in disruptive research may accrue to a diffuse group of public and private stakeholders, the broader benefits – regional economic clustering, industrial agglomeration, knowledge spillovers and national strategic advantage – are substantial and long-lasting.

Vast new frontiers of progress lie before us. Emerging platform technologies – particularly AI – will transform every field of research, as well as the very foundations of scientific inquiry. The scale of transformation over the coming decades is likely to eclipse that of the post-war period, with “intelligence too cheap to meter”, programmable life forms and many other revolutionary capabilities as yet unimagined.

For the UK to play a central role in shaping future technoscientific revolutions, it must confront a difficult question: is today's institutional landscape best placed to foster and capitalise on the kind of work that sparks them?

Here, there is reason for concern. The last time British public-research institutions assumed a leading and lasting role at the forefront of a disruptive, industry-creating field was in the 1950s and 60s, with the establishment of the LMB. In the decades since, the UK has struggled to replicate this success. Only DeepMind, which is funded by a US tech giant, has achieved comparable global influence and centrality within a transformative new field. The UK continues to produce many isolated standout breakthroughs, but it has not created environments that sufficiently concentrate the interdisciplinary talent and resources needed to develop new discoveries, inventions and ideas into pioneering fields and industrial ecosystems.

Developing a strategy to correct the course is no simple task, and no single template will suffice. However, there is guidance: a substantial body of work examines how new domains take shape, and what institutional conditions help them scale. Next, we distil this literature into frameworks for understanding the origins of technoscientific revolutions. We then turn to the emerging field of applied metascience, which seeks to create new kinds of research organisations that put these principles into practice.

An Outline of Disruptive Research and Technology

How are new scientific fields born? One of the most oft-cited frameworks is that of the “paradigm shift” in scientific understanding, as described in Thomas Kuhn's 1962 work *The Structure of Scientific Revolutions*. Kuhn contends that science does not progress solely through the incremental accumulation of knowledge within accepted frameworks, which he refers to as periods of “normal science”, but also through occasional revolutions in which anomalies accumulate, the old framework fails and a new paradigm

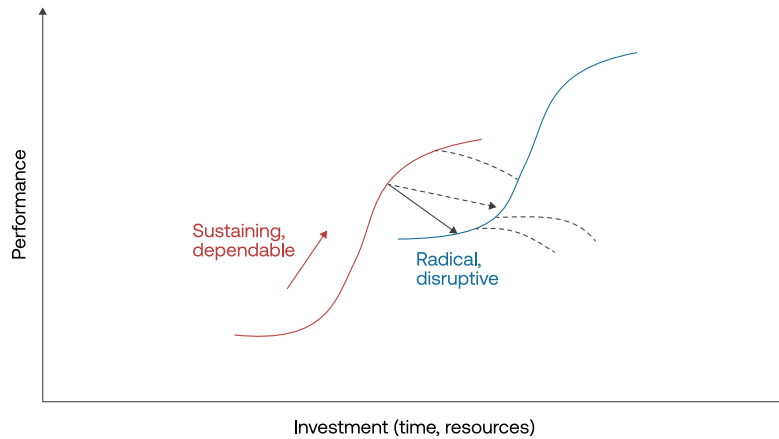
emerges to redefine the landscape. In this view, certain discoveries or inventions create what are, in effect, new domains of science, overturning core assumptions of the old order.

These sorts of paradigm-shifting breakthroughs can be compared to the “heroic” science outlined by Brenner. He describes three phases of scientific research: the heroic, which pioneers new fields; the classical, which expands and organises them; and the bureaucratic, which consists of routine tasks such as the assembly of data sets.⁵⁷ Other thinkers such as Imre Lakatos, Paul Feyerabend and Michael Polanyi offered differing accounts of scientific progress, but their collective work highlights the qualitative differences between research that consolidates and research that transforms – and outlines that different institutional conditions are suited to each.

These frameworks – along with others including economist Clayton Christensen’s work on disruptive innovation – have been highly influential in shaping the strategies of some of the world’s most successful research funding agencies, such as the Defense Advanced Research Projects Agency (DARPA) in the US, which presents the diagram below to illustrate how the interplay between incremental and disruptive work can drive growth. While disruptive innovations initially may not fit into or improve upon current methods or paradigms, if nurtured, they can eventually displace established models. Many others have advanced related arguments.^{58,59,60,61}

FIGURE 1

A model for disruptive innovation



Source: DARPA

At the heart of these theories of progress are two core concepts:

- Disruptive research has historically resulted in transformative leaps, and a meaningful distinction can be made between disruptive (transformative) work and work in already mature fields.
- Actively pursuing and supporting disruptive work requires alternative organisational setups and incentives to those that might best support large-scale work in mature fields.

We examine each of these in turn.

MATURE AND DISRUPTIVE RESEARCH

We begin with the distinction between mature and disruptive research.

- **Mature research** is consolidative and iterative, building upon and consolidating existing streams of knowledge to expand our understanding within a field (or across fields). It develops established frameworks or improves upon existing types of technology.

- **Disruptive research** is transformative in nature, representing the introduction of an entirely new framework of ideas that changes the context in which other work is done. Disruptive discoveries or inventions often displace established knowledge, methods and theories to push science in a new direction.

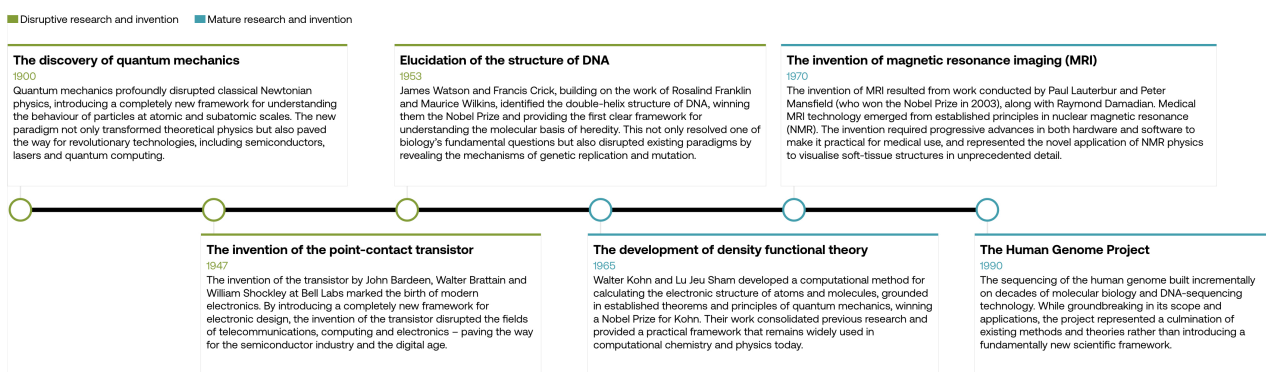
Put simply, disruptive work often involves exploration at the highest conceptual level, creating new paradigms rather than operating within existing ones. Both types of research are essential for a dynamic scientific ecosystem, and the interplay between the two propels progress. Disruptive breakthroughs provide the leaps that redefine what is possible, while mature research expands, refines, translates and scales – deepening understanding and realising the practical benefits of those leaps.

It is important to note that disruptive research is not synonymous with curiosity-driven or blue-sky research. A curiosity-driven programme can exist entirely within an established theoretical framework without challenging fundamental theorems or approaches, while a disruptive breakthrough can emerge from either curiosity-driven or goal-oriented efforts. Fostering disruptive research therefore requires a different approach to simply championing blue-sky initiatives.

The following examples demonstrate the distinction between mature and disruptive research, showing how each has yielded major advances in science and technology.

FIGURE 2

Both disruptive and mature research in science and technology have led to critical advances



Source: TBI

New scientific opportunities that spring from disruptive breakthroughs must be exploited through conventional research efforts, so a dynamic ecosystem must be capable of supporting both. Likewise, a robust R&D strategy should strike a balance between support for science aimed at advancing knowledge, skills and products within existing fields, and support for disruptive science to seed the industries of tomorrow. Increasingly, researchers and metascience thinkers argue that, in recent decades, the R&D systems of scientifically advanced countries have inadvertently inhibited disruptive science and that, as outlined in recent analysis by academics James Evans and Johan Chu, “fundamental progress may be stymied if quantitative growth of scientific endeavors ... is not balanced by structures fostering disruptive scholarship”.^{62,63,64}

AN INSTITUTIONAL GAP FOR DISRUPTIVE DISCOVERY AND INVENTION

The conditions that encourage the emergence of new domains often differ profoundly from those that sustain mature ones. However, because most research activity naturally occurs within established fields, ecosystems tend to optimise for more predictable, incremental progress – inadvertently creating environments that crowd out disruptive work.

Any attempt to nurture a specific form of research should begin with one fundamental maxim: different types of research demand distinct institutional structures. In business, it is well understood that organisational design and incentives must align with purpose; similarly, the high-level structure of a scientific institute should be tailored to its mission.

Key dimensions of variation that can influence the capacity of a scientific institution or programme include:

- **Management styles.** Some research efforts thrive under active, milestone-oriented management, while others require autonomy and a light-touch approach to encourage independence and intellectual freedom.
- **Time horizons.** A group oriented towards 20-, 30- or 40-year impact will behave very differently to one aiming to demonstrate the value of its work in less than five years.
- **Funding models and incentives.** Milestone-oriented, project-based grants are the dominant mechanism for the financing of large-scale mature research, whereas unconstrained, long-term funding is often necessary to support high-risk exploration.
- **Technology and infrastructure.** The degree to which technical support and equipment are centralised and easily accessible has marked effects on research culture.
- **Governance and risk tolerance.** High-risk, high-reward work requires a governance model that tolerates a level of uncertainty and failure that may be unsuitable for investment-heavy, large-scale science and invention.

- **Internal hierarchies and career structures.** Flat hierarchies foster open exchange and creative exploration, and are a better fit for exploratory work. More structured hierarchies, with clearly defined and specialised roles, can enhance stability and efficiency in large-scale, goal-oriented projects.

Each of these elements shape how research organisations function and influence their capacity to stimulate different types of progress. With so many axes of variation to explore, the scope for reimagining models of R&D funding and governance, and the structuring of research institutes, is vast. However, this potential “design space” remains largely unexplored in the UK, where research organisations share a remarkable degree of homogeneity in their core organisational models.

A degree of diversity in form, mission and culture can be found within UK universities and other organisations at the level of labs, buildings and departments, and universities have historically made significant contributions to disruptive work. However, they typically exist within broader institutional frameworks that impose immutable fixed features, creating systematic barriers to pursuing specific types of disruptive technoscience at scale. These include peer-reviewed publication, project-specific grant funding, university and funder bureaucracy and, in many cases, teaching responsibilities. Because universities and many other public-sector research institutions are tied to the existing academic career path, they inherit the same set of incentives. We briefly survey the existing landscape in the box below.

Disruptive research and associated constraints in the existing institutional landscape

Universities: Universities have played a crucial role in the transmission of knowledge and served as spaces for unrestricted intellectual exploration. They have historically seeded many disruptive discoveries, from penicillin to early molecular biology, and remain central to the UK's research ecosystem, offering unparalleled breadth across disciplines. However, with breadth comes a weakness. Their wide array of large-scale activities – teaching, knowledge transmission, the bulk of mature research – means they are prone to excessive bureaucracy, additional demands on researchers' time, and are pressured to maximise various metrics on short-term timelines (such as citations or research “impact” assessments).

Decentralised units: At “hub and spoke” institutes such as the UK Dementia Research Institute, a central organisational structure supports multiple distributed research groups, often embedded within different university departments. This model leverages existing university infrastructure but fragments teams, both physically and organisationally. Researchers may be nominally part of a shared institute but in practice remain embedded in the wider academic system, where they are subject to the same incentives, career pathways and constraints.

Standalone institutions: These institutions have more focus than universities and bring a high density of talent to work collaboratively on a specific area. One example is the European Molecular Biology Laboratory which, through its partnership with Google DeepMind, helped to create AlphaFold. Such institutions can also house offshoots with the potential to grow into new fields; a good example is the LMB's support of Jason Chin

and his laboratory's work on reprogramming the genetic code.⁶⁵ However, the current landscape of these institutions reflects already relatively mature (though still evolving) fields of science, such as those of the Wellcome Sanger Institute and today's LMB. In addition, many of these institutions adopt an academic career path and structure, unlike the majority of private-sector labs such as DeepMind; consequently, they largely remain inside the incentive structures of academia. One UK-based exception is the London School for Mathematical Sciences (LIMS). It is the country's only independent research institute in theoretical physics and mathematics, where researchers helped initiate and progress the field of AI-assisted discovery in maths and physics. Founded as an alternative to universities, LIMS has taken more decisive steps away from academic structures, operating with no traditional research groups or principal investigators.

Public-sector research establishments: The STEP programme at the Culham Centre for Fusion Energy sits at the global forefront of fusion technology, pursuing a major industry-scale programme. And during the Covid-19 pandemic, Porton Down's rapid response in assessing the performance of lateral-flow tests enabled the UK to be the first country to embrace the new paradigm of rapid asymptomatic testing at scale. However, a large majority of these organisations have well-defined and mature missions. While they can contribute to creating and scaling new fields of research, it is not their primary objective.

The ongoing AI revolution demonstrates the effects of these constraints. UK public-research organisations, though vital sources of talent, have generally not driven frontier advances to any significant extent. Existing public institutions have struggled to compete with tech-funded labs which,

unbounded by academic constraints, have driven much of the most fundamental progress, from transformers and deep reinforcement learning to instantiation in tools such as GPT-4 and AlphaFold.

The key question facing the UK today is this: can a critical concentration of opinion coalesce around organising principles for new ways of pursuing technoscientific progress, which could also enhance the capabilities of existing institutions?

Applied Metascience and New Research Models

The applied metascience movement represents a response to this need. It seeks to unlock the latent potential of research by building new scientific structures and institutions, with a focus on enabling researchers to pursue disruptive and impactful work.

Historically, the creation of new institutions of science has been an immensely powerful way to accelerate progress – with the most significant institutions arguably as influential as any scientific discovery or even field. From the creation of the Royal Society, through to the emergence of major industrial research labs in the 20th century and entities such as the US's Advanced Research Projects Agency (then ARPA, now DARPA), research organisations and funders whose structures were tailored to specific visions or missions have reshaped the trajectory of human progress. These new kinds of institutions acted as disruptive inventions in and of themselves – and this unlocked step-changes in discovery, invention and impact.

Over the past several decades, scholars of science and technology have produced a rich body of work examining the history, structure and inner workings of research institutions. Yet despite these insights, the creation of entirely new entities – on the scale of the Max Planck Society or the US national labs – has remained outside the realm of practical discussion in the UK. However, a window of opportunity may have opened. Recent years have seen a resurgence of interest in reimagining the structures of science in this country, driven by a convergence of factors: the growing influence of technology-focused philanthropists, renewed government attention on

national research capabilities and increasing recognition among researchers that the existing systems do not empower them to make the best use of their talents.

The call to build was put best by Alan Kay – Xerox PARC researcher and pioneer of personal computing – who said, “The ARPA/PARC history shows that a combination of vision, a modest amount of funding, with a felicitous context and process can almost magically give rise to new technologies that not only amplify civilization, but also produce tremendous wealth for the society. Isn’t it time to do this again by Reason, even with no Cold War to use as an excuse?”

This recent surge of interest has fuelled the beginnings of a movement to turn these ideas into reality at scale. The applied metascience movement, unlike traditional efforts at academic reform, takes a creative, entrepreneurial approach to institutional design, informed by a mix of historical, social and technical analysis. It seeks not only to tweak existing models but to invent new ones, operating from a “blank slate” assumption about how research should be organised. Recent successes have been featured in *The Wall*

Street Journal, *The Atlantic* and *Nature*.^{66,67,68}

This endeavour, in the context of life sciences, was recently reviewed in the prominent journal *Cell* by Sam Rodriques, himself a founder of two new institutional models.⁶⁹ In *The Institutional Menagerie*, Rodriques explores the plethora of new research models arising (almost exclusively in the US), including the Covid Fast Grants scheme, FROs, Astera, the Arc Institute,

Arcadia and Future House.^{70,71,72}

The UK government has been a leader in this space. While Silicon Valley and Boston have led in terms of the number of organisational models deployed, this has been through philanthropic rather than public money; the UK government has been alone in explicitly creating new entities to pursue a metascience agenda. This has primarily been through the Advanced Research and Invention Agency (ARIA), launched in early 2023. While the Vaccine Taskforce and AI Security Institute have represented innovations in

the interaction between government and technology, ARIA is the first government agency with the explicit legal freedom to innovate in terms of how science is conducted and funded.

ARIA is modelled in part on the early ARPA of the 1950s and 1960s, with wide-ranging freedom to choose its own priorities, and is free from many typical bureaucratic constraints; programme directors are given an unusual degree of personal autonomy to manage research funding portfolios, enabling greater risk-taking. Though it is early days, it is already funding a range of cutting-edge programmes, from neurotechnology to controlling advanced AI systems and exploring new modalities of computation. It has received prominent positive coverage,^{73,74} continued support on a cross-party basis, and has first-rate scientists and technologists on its board and in its advisory group, including Nobel laureates David MacMillan and Demis Hassabis.

ARIA represents a bold step forward for UK innovation, but its work must be complemented by new types of physical research institutions. While ARIA's model is excellent for identifying and funding promising, ambitious, high-risk projects, it relies on existing academic, public or private research institutions to carry out this work. And though ARIA does see a strong need to catalyse new institutions, and has been effective thus far at doing so, it is not in a position to create and sustain a network of physical institutions on the scale of our proposal.

A gap remains in the UK's research landscape for physical institutes that bring together a critical concentration of creative researchers to work collaboratively on nascent areas of technoscience over a sufficiently long period of time – an ARPA for physical labs.

Our proposal draws inspiration from this growing international ecosystem of institutional experimentation to put forward a distinctly UK-oriented model, focused not just on funding mechanisms but on building enduring physical environments tailored to the needs of disruptive invention.

The aim of TBI's proposal

To attract a critical concentration of diverse talent to work together side by side on a nascent technoscientific window of opportunity, in a manner spanning engineering, discovery and theory. This would be at a stage when the private sector is unlikely to see a clear opportunity, but which could generate major spillover effects and invite future private investment within a decadal timescale.

This aim captures how some of the most transformative laboratories of the 20th century operated, and how the likes of DeepMind operate today. In the next section we explore the striking organisational commonalities shared between these landmark labs – which stand in stark contrast to conventional wisdom regarding how public research should be organised – and demonstrate how these common features facilitate the pursuit of disruptive work.

03

Lessons From Transformative Labs

Actively pursuing disruptive research means cultivating environments suited to this purpose. This section explains why such environments must depart significantly from today's academic norms. Contemporary structures –which often emphasise short-term outputs, divide basic from applied research, and rely on external assessment and funding through peer review – tend to favour consensus-driven ideas and block rapid cycles of discovery and invention. As a result, they risk systematically suppressing the conditions under which disruptive work tends to emerge.

To understand how things might be done differently, below we outline a different approach to research – one that treats discovery and invention as tightly coupled, and therefore calls for a different kind of physical lab. We then look to historical labs that redefined frontiers of science and technology, comparing them to present-day norms. We focus on three preeminent 20th-century institutions: Bell Labs, Xerox PARC and the early LMB. Each eschewed many of the deeply ingrained assumptions that have characterised public R&D since the end of the second world war, sharing a set of organisational principles that allowed them to pioneer technoscientific revolutions in their domains. We close this chapter by briefly evaluating more recently established organisations based upon similar principles, including DeepMind and the Arc Institute.

Cycles of Discovery and Invention

Those immersed in the day-to-day practice of science often take the high-level structures of their lab, institution and ecosystem as a given, especially if they have been immersed in them for a long time. Yet these systems are not fixed; they are the result of a mix of deliberate top-down policymaking, decentralised revision and drift, and historical inertia.

Today's R&D systems are largely rooted in assumptions forged in the aftermath of the second world war, which triggered a reforging of the relationship between science and the state. Recognising the critical role that

science and technology had played in securing the Allied victory, President Franklin D Roosevelt tasked the engineer Vannevar Bush with designing a peacetime strategy to harness research for national progress. Bush's report, *Science: The Endless Frontier*, laid the foundations of a post-war research model that continues to shape Western R&D policy today.⁷⁵

At the report's heart is a division between basic and applied research, with Bush writing of the need to "increase the flow of new scientific knowledge through support of basic research" to bolster industry. It argues that the state should support the generation of knowledge through basic science, with industry then picking up the baton and developing that knowledge into commercially valuable innovations – the so-called "linear model" of R&D.

These assumptions are now so deeply entrenched in contemporary R&D policy that they are rarely questioned or even acknowledged. For example, the 2015 report that led to the creation of UK Research and Innovation (UKRI) echoes the separation between basic discovery science and practical application, confirming it as a primary division in research.⁷⁶ Similarly, Germany's Max Planck Society and Fraunhofer Society institutionalise a separation between the pursuit of knowledge and its practical application.

There are scenarios in which this model captures the nature of research well. However, many scholars have argued that the linear model fails to capture the complex interplay between scientific and technological progress – and between basic and applied research. American political scientist Donald Stokes, in his book *Pasteur's Quadrant*, highlights "use inspired" basic research that seeks fundamental understanding and immediate utility.⁷⁷ Science-policy scholar Michael Gibbons and colleagues echo this sentiment in their description of "mode 2" knowledge production, in which knowledge is created in the context of application by transdisciplinary teams, rather than within isolated academic disciplines.⁷⁸ Others – including innovation researchers at the Science Policy Research Unit and evolutionary economists such as Giovanni Dosi similarly argue that scientific advances and technological inventions co-evolve in practice, reinforcing each other through cycles of feedback and refinement.^{79,80}

In their 2016 book *Cycles of Invention and Discovery: Rethinking the Endless Frontier*, former Bell Labs department head Venkatesh Narayanamurti and science and technology scholar Toluwalogo Odumosu outline a version of this cyclical model in which discovery and invention are deeply interdependent and mutually reinforcing processes, with close feedback loops driving rapid progress.⁸¹ They argue that breakthroughs often emerge where the two converge, especially in physical spaces where tools and theories evolve together. A subsequent 2021 book by Narayanamurti and physicist Jeff Tsao, *The Genesis of Technoscientific Revolutions*, further develops this theory.⁸²

This body of work suggests that new domains are catalysed by hybrid technoscientific efforts that integrate fundamental enquiry with practical problem-solving, tool-building and invention. In labs that embody this approach, diverse teams made up of scientists, engineers, theorists and technicians test and discard hypotheses, build and dismantle prototypes, and rapidly iterate on tools and ideas. But despite their disproportionate impact on technoscientific progress, such labs remain rare in national R&D systems. A global institutional gap for a similar sort of research environment was highlighted by Ilan Gur in 2020, prior to becoming the first CEO of ARIA.⁸³

Physical Proximity Is Everything

For cycles of invention and discovery to take root, proximity matters. The iterative exchange between ideas and tools, theory and application, is most powerful when researchers are brought together in a shared physical space. Notable examples of such environments include Building 20 at the Massachusetts Institute of Technology (MIT), Skunk Works, Bletchley Park, Bell Labs, Xerox PARC and Los Alamos – where the collision of minds and ideas sparked chain reactions.

Empirical studies corroborate that geographical proximity enables unconventional innovation^{84,85} yet the true power of co-location may be best understood intuitively: people often create visions together, explore them together and actualise them together. Shared physical spaces

cultivate a communal genius, where ideas – iterated upon and amplified within a supportive and dynamic community – flourish more fully through a collective social feedback loop.⁸⁶

Several features of these hotbeds of creativity cannot be replicated online or through infrequent real-world collaborations:

- **Shared physical infrastructure.** The use of shared resources, especially important in the physical sciences, negates the need for individual researchers to have their own equipment, which can be very expensive. This in turn reduces the need for the very large professor-led research groups that have become prominent in academia.
- **Close feedback loops and integrated teams.** Researchers working in edge-of-the-art spaces where many experiments fail must compensate through volume, creativity and persistence. The rate of turnover of new ideas, new approaches and new methodologies is very high; to quote US chemist Linus Pauling, “The best way to have a good idea is to have a lot of ideas.” This dynamism requires equally accelerated feedback loops, integrating input from engineers, technical staff and scientists with different specialties. The ability to pop down the hall to ask a quick question should not be underestimated, nor the ability to quickly share reagents or help conduct one another’s experiments.
- **Informal cross-pollination.** Proximity generates “idea collisions”, whereby unconnected ideas collide to create something new. The more that researchers share or pass through a physical space and the greater the amount of time they spend communicating, the higher the frequency of these unplanned productive collisions.⁸⁷ Researchers who might not have actively sought each other out as collaborators could meet by chance, with the resulting cross-pollination sparking new directions for each.⁸⁸
- **Tacit knowledge.** Much of the value of unplanned interactions in hallways, cafeterias and labs lies in the exchange of tacit knowledge – knowledge that doesn’t tend to come up in formal written or verbal communication. An analysis of research papers and patent applications from the past 50 years found that teams working remotely were less likely to produce disruptive discoveries than on-site teams. It also found that

remote communications often focus on late-stage technical tasks with more codified knowledge, and are less commonly used for workshopping half-formed ideas or conceptual tasks that involve tacit knowledge.⁸⁹

- **Inspiration.** Physical spaces can create a unique energy. Something akin, as Mervin Kelly said of Bell Labs, the organisation he presided over, to being part of a living organism of aligned scientific discovery.

All modern research spaces should make use of the best available online resources to collaborate widely and leverage the manifold benefits of the digital age. But online spaces cannot replicate the spontaneity and richness of a shared real-world environment. Digital platforms are excellent for refining ideas, but they fall short in the messy, creative early stages of equipment-based science, where ambiguity reigns and serendipitous connections are invaluable.

The journalist and author Jon Gertner summed it up best when he wrote of Bell Labs that “physical proximity was everything”.⁹⁰

Lessons From Disruptive 20th-Century Labs

The cyclical model of research found some of its most compelling expressions in a group of revolutionary 20th-century labs.

The impact of AT&T Bell Labs, Xerox PARC and the early LMB as engines of invention is difficult to overstate; each was amongst the most impactful drivers of 20th-century technoscientific progress. Their work laid key foundations of personal computing, molecular biology and communications technology, seeding international networks of science with these founding institutes at their core. It is notable that, when considering each of these three revolutions, such singular labs played such disproportionate roles – a feature seen more recently with AI.

Model Labs and Their Impact

AT&T Bell Labs (1920s to 1990s)

“My first stop on any time-travel expedition would be Bell Labs in December 1947.” Bill Gates⁹¹

Few institutions in history have had as profound an impact on technological progress as this much-studied US-based industrial research organisation. Between its founding in 1925 as the research arm of AT&T and the beginning of its breakup in 1984 following an antitrust lawsuit, researchers conducted work that would lead to ten Nobel prizes and four Turing Awards – and filed on average almost 400 US patents each year.^{92,93} Though it originally focused on telecommunications technology, the impact of the discoveries made at Bell Labs continues to reverberate through disparate sectors. Its principal innovations – including vacuum tubes, the transistor, fibre optics, radio astronomy, UNIX and C programming, and information theory – enabled the miniaturisation of technology and revolutionised global communications.⁹⁴

Xerox Palo Alto Research Centre (early 1970s)

“I was so blinded by the first thing they showed me, which was the graphical user interface. I thought it was the best thing I’d ever seen in my life.” Steve Jobs

Xerox PARC, a small corporate research centre, is widely credited as blazing the trail that resulted in the modern personal computer, laying the foundation for today’s digital age. Initially building on ARPA’s computing program, PARC’s inventions – from the graphical user interface, the mouse

and Ethernet networking to object-oriented programming and laser printing – defined the user experience and infrastructure of modern computing. Researchers have won four Turing Awards for work conducted at the institute in the early 1970s, from a core research staff of barely 50. It was, as M Mitchell Waldrop called his book, “The Dream Machine”.⁹⁵ Despite its small size, PARC’s legacy endures, credited by Bill Gates and Steve Jobs for dramatically accelerating the personal-computing revolution and influencing the development of Windows and the Mac respectively.^{96, 97}

The MRC Laboratory of Molecular Biology (1950s to present)

“The most productive centre for biology in the history of science.” James Watson⁹⁸

The LMB, officially founded in 1962 on the outskirts of Cambridge, England, grew from the Medical Research Council (MRC) Unit for the Study of the Molecular Structure of Biological Systems (established in 1947) and is one of the most influential institutes in the history of biological research. Unlike Bell Labs and PARC, which were the research arms of large corporations with effective monopolies, the LMB was backed exclusively by the MRC (a public body) to pioneer the emerging field of fundamental molecular biology. The genesis of the institute can be traced to a bold decision by the MRC to fund an unorthodox unit of researchers led by Max Perutz, John Kendrew, Aaron Klug and Fred Sanger, already at that time a Nobel laureate. It has since produced 12 Nobel prizes, including four laureates in its first year.⁹⁹ By this imperfect measure it is the most successful scientific research institute in history. Originally focusing on three themes (structural studies, molecular genetics and protein chemistry), the breadth of research at the LMB has since expanded, along with the institute itself. Scientists in the LMB’s early days unravelled the structure of DNA, developed protein- and genetic-

sequencing technologies and pioneered the use of X-ray crystallography and electron microscopy to determine protein structures, along with many more discoveries that collectively established molecular biology as the bedrock of modern medicine.

These exceptional institutes were, as Bell Labs' president Kelly said of his own facility, no houses of magic. Despite their independent foundations and differing focuses, they converged on a strikingly similar set of structural and cultural principles that made them fertile ground for transformative research. Other pioneering labs, such as Lockheed Martin's Skunkworks, shared similar features.¹⁰⁰

Through examination of these commonalities it is possible to derive guidance for designing the institutes of tomorrow. While we would not suggest attempts to slavishly copy the 20th-century labs, we hold them up as "existence proofs" of a highly productive model for enabling scientific talent – one that is largely absent from the modern UK R&D ecosystem and public R&D globally.

Core Elements for Building Breakthrough Labs

From these model labs, we distil three broad design principles that enabled disruptive technoscience to thrive and scale.

1. **Focus on a scientific vision.** Rather than organising their efforts around a discipline, in the manner of modern university departments, or focusing on a specific goal (such as mapping the human genome), these institutes were physical labs collectively focused on broad visions for the future of

science or technology. At the time of their inception these visions were not established fields of research – they more closely resembled heretical bets against mainstream opinion.

2. **Protection from external pressures.** Researchers were evaluated internally by those familiar with their project, their potential and their body of work. Avoidance of external evaluation was possible because the institutes benefited from full, internally distributed core funding, derived from a single source. The insulation of individual researchers from external evaluation is key: research contrary to established opinion is more likely to be penalised or misunderstood by external peers. Researchers were not guided by the need to constantly churn out papers, as they did not apply for grants, and so were freed from the associated bureaucracy.
3. **A community approach.** The internal allocation of power and resources is a significant determinant of how an institute functions and what kind of science it can pursue. These institutes were characterised by flattened hierarchies, small teams that collaborated dynamically and organically, light-touch management and strong technical core facilities and staff. The notion of separate internal “labs”, such as in university departments, was largely absent – as was the modern phenomenon of a single principal investigator (PI) putting their name on the ideas and publications of dozens of “trainees”. Those presenting research conducted it with their own hands. They resembled a society of fellows, rather than a conventional academic hierarchy. Labs were evaluated as a whole, not at the level of individual researchers.

We now examine each of these.

1. FOCUS ON A SCIENTIFIC VISION

Each institute championed what can be described as an overarching research vision. It is difficult to concisely define what constitutes a scientific or technological “vision”, but this does not render the concept meaningless. It is not a field, a discipline, a plan for a specific invention nor, in most cases, a well-defined ultimate goal. It does represent a new and often outlandish view of a scientific horizon. As the quantum scientist and applied metascientist Michael Nielsen explains, a researcher presenting a vision is

“effectively a would-be prophet standing up and proclaiming: ‘I see a wonderful opportunity over there, that looks [something like this]. Let’s go explore!’”¹⁰¹

Kay, of Xerox PARC, wrote of the importance of such visions. “[A] great vision acts like a magnetic field from the future that aligns all the little iron-particle artists to point to ‘North’ without having to see it ...The pursuit of Art always sets off plans and goals, but plans and goals don’t always give rise to Art. If ‘visions not goals’ opens the heavens, it is important to find artistic people to conceive the projects.”

Model Labs and Their Visions

Bell Labs: In 1907, AT&T put forward a motto: “One policy, one system, universal service.” This phrase encompassed Bell Labs’ overarching vision of unifying telecommunications to offer global connectivity to its customers. Bell Labs operated under the assumption that the telephone system should be an integrated, seamless network, and its scientists were given remarkable freedom and long-term funding to pursue research that might one day serve this mission. Bell Labs exemplified how a technically coherent long-term goal, paired with patient institutional support, can generate innovations decades ahead of their time.

Xerox PARC: According to the vision of American computer scientist Joseph CR Licklider, PARC was established to develop “the architecture of information” and operated on the belief that “computers are destined to become interactive intellectual amplifiers for everyone in the world,

universally networked worldwide”. This vision stood in stark contrast to the dominant paradigm of the time, which imagined computing as the domain of massive mainframes used by specialists via punch cards.

LMB: The guiding vision of the early LMB was that a true understanding of cellular life would come from comprehending the interactions of individual molecules – in particular the precise three-dimensional structures of proteins, nucleic acids and their assemblies. This approach, deeply structural and reductionist, was radical at a time when much of biochemistry focused on bulk-tissue studies, cataloguing metabolic reactions or measuring enzyme activity without a mechanistic or structural foundation. The LMB’s founders believed that biology could become a fundamental science only by rooting itself in the physical sciences: physics, chemistry and crystallography – which transformed biology from a largely descriptive field into a mechanistic one. This laid the foundations for molecular biology, biotechnology and much of modern medicine.

Today the visions that once guided these institutes may seem self-evident, as though their success was always assured. But at the time these visions were radical, existing at the margins of accepted thinking in their respective fields. In practice, institutes destined to become groundbreaking often pursue “weird” ideas that are either unpopular or largely unintelligible to the wider research community. The ambitious, unorthodox visions uniting places such as Bell Labs, Xerox PARC and the LMB allowed them to become global focal points for talent – they became “the place to be” to work on their topics.

Compared with FROs, vision-oriented institutes are somewhat broader in scope and duration, while also being more open-ended in how success is defined. These institutes are shaped less by a specific technical target and more by a shared sense of possibility. Yet their research differs from blue-sky or purely curiosity-driven research in that it is cohesive and directed; not a loose federation of individuals or labs working entirely independently, but a collective journey towards an uncertain frontier. This section explores how such visions shape breakthrough science, and why they matter for institutional design.

Vision-Oriented Research Helps Avoid Field Stagnation

Scientific and technological fields are predisposed towards incremental progress, increasing specialisation and the pursuit of refinements within a dominant paradigm. This is not inherently problematic – indeed, much cumulative progress depends on it – but it can create a kind of intellectual inertia. Over time, fields often drift into local optima: zones of intense activity and apparent success that, while productive in the short term, obscure radically better possibilities that lie outside the prevailing conceptual framework.

This tendency is reinforced by the structure of modern academia, where research careers are often built on obtaining reliably publishable results that extend existing theories or technologies in peer-acceptable ways. Fields can become blind to alternative starting points – or even to the idea that such starting points exist.

A well-articulated scientific vision, meanwhile, functions like a north star: it does not prescribe specific outcomes or step-by-step plans, but illuminates an alternative horizon. It invites researchers to explore conceptual terrain that is currently invisible or considered fringe from within the dominant paradigm. In doing so, it creates space for transformative ideas that could not emerge from the logic of stepwise progress.

The rise of personal computing is one such example: it did not emerge from incremental improvements to mainframe systems because it required a wholesale shift in assumptions – a shift enabled by a vision of what

computing could become, and by the institutional backing to pursue it. It was difficult to justify this vision to the wider field at the time – only a few understood the opportunity and pursued it.

CASE STUDY

The Genesis of Personal Computing

Computing was on a different trajectory before the interventions of the ARPA computing program and Xerox PARC. In the 1950s, the familiar form of a computer was a vast mainframe machine that occupied most of the room. Users provided commands within the confines of a reasonably limited predetermined architecture and returned when the computational functions were completed. The default path for the development of computers was broadly to continue in this vein – perhaps increasing machine size, ramping up processing speed or investing in other incremental improvements.

Instead a revolutionary 1960 paper by Licklider, “Man-Computer Symbiosis”, presented a radically different vision for computing – one that would ultimately become the bedrock of Xerox PARC.¹⁰² He envisioned computers not as remote, impersonal, mainframe machines, but as portable, accessible, interactive tools capable of working in real-time alongside humans, augmenting their intellect by facilitating problem-solving and idea formulation in a deeply integrated, symbiotic partnership. This vision was astonishingly bold in its optimism and foresight, challenging every prevailing assumption about what computers could become.

Xerox PARC won three Turing Awards for its work in the early 1970s, in large part guided by Licklider’s vision, and with a core team of barely 50 researchers. More importantly than prizes, it created personal computing.

Focus on a Broad Vision Allows Researchers to Pivot to Surprise

Visionary framing does more than set direction: it creates the conditions for agility. When researchers are united by a broad, generative vision rather than pre-defined milestones, they are free to pivot to surprise and can shift course in response to the unexpected.

Historically, such moments of serendipity have launched transformative research trajectories. But capitalising on them requires research environments that prioritise responsiveness, tolerate failure and protect researcher autonomy. As physicist Tsao, co-author of *The Genesis of Technoscientific Revolutions*, observes, in rigid and prescriptive systems – where researchers are pressured to meet predefined objectives or timelines – these sparks of discovery can be quickly extinguished. Today’s academic ecosystem can struggle to support such flexibility. Grant-funded researchers can lack the freedom – or the incentive – to pursue unplanned observations when such insights lead them outside the scope of their project timelines or disciplinary boundaries.^{103,104} This is especially true for PhD students and postdoctoral researchers (who make up the majority of the research workforce) should they lack support from their PIs.

The tension between short-termism and the freedom to follow emerging insights leads to a further point: visionary science thrives not just on autonomy, but on time. The most transformative breakthroughs often come not from chasing immediate impact, but from pursuing long-range missions that leave room for exploration, failure and surprise. The guiding visions of the three model institutes were oriented towards 20- to 30-year horizons – allowing researchers to venture deeply into the unknown – yet each delivered remarkable advances far sooner.

Bell Labs operated on a 30-year timescale from invention through to application,¹⁰⁵ with founder Theodore Vail promoting a long-term strategy measured “in decades instead of years”.¹⁰⁶ At Xerox PARC, Kay’s role in the personal-computing revolution was to imagine what could be possible in computing in 20 to 30 years’ time, and begin the journey to that summit. At the unit that grew into the LMB, future Nobel laureate Brenner, on seeing a

preview of Francis Crick and James Watson’s double-helix model, caught the first glimpse of a huge new field on the cusp of creation, and set about the decades-long task of bringing it into being.

We next turn to an exploration of how these communities – pursuing research visions often at odds with prevailing wisdom – were protected from the external pressures that characterise today’s academic system.

2. PROTECT RESEARCHERS FROM EXTERNAL PRESSURES

The emergence of revolutionary work at the three institutions highlighted depended on environments purpose built to sustain long-range thinking, protect exploratory work from short-termism and shield scientists from the pressures that dominate much of the public R&D landscape today. They benefited from generous core funding, internal assessment by experts familiar with the research and the deliberate minimisation of teaching, managerial and administrative responsibilities. Researchers focused fully on advancing their ideas, unencumbered by the incentives and constraints that shape academic careers.

It is understandable – necessary perhaps – that large-scale business-as-usual research necessitates a lower tolerance for risk; as such, greater levels of oversight and increased bureaucratic complexity are inevitable.¹⁰⁷ Nonetheless, today’s public R&D ecosystem is replete with constraints on scientists’ time, budgets, and research programmes – generating widespread dissatisfaction, limiting ambition and diluting focus on the core objective of doing good science.¹⁰⁸

“Hire the brightest scientists possible and let them determine what to do. They are better placed than anyone to work out where the opportunities lie”

Historically transformative institutes created fertile environments around exceptional researchers, removing all manner of barriers. As Hugh Pelham, former director of the LMB, observed, “The principle established early on was that the best way to make breakthroughs is to hire the brightest scientists possible and let them determine what to do. They are better placed than anyone to work out where the opportunities lie, and some of the greatest discoveries come from unexpected angles.”¹⁰⁹

What follows are specific design features that allowed researchers to pursue disruptive ideas by shielding them from both internal and external pressures.

Minimal Distracting Responsibilities

The protective environments of the model labs allowed almost exclusive focus on the science, with little to no time spent on formal teaching or writing grant applications.¹¹⁰ For today’s researchers, in many fields, ascending the academic career ladder forces them away from the very activities that drew them to science: experiments at the bench and self-directed discovery. Postdoctoral scientists who become PIs may gain the freedom to set their new lab’s research agenda, but they also assume the roles of an administrator, manager and fundraiser responsible for a team of researchers. In most cases there are also teaching obligations, with accompanying administrative and preparatory tasks often taking three or four times as long as the teaching itself.

This arrangement can leave only brief interludes in researchers’ careers where they can engage in entirely self-directed experimental science – uninfluenced by the research programme of the lab’s PI in whose lab they work, but also unhindered by additional responsibilities. According to a 2020 study by the Wellcome Trust, only 45 per cent of UK researchers surveyed

felt able to effectively balance the competing roles required as part of their employment, while 44 per cent reported that taking on numerous roles left them little time for research.^{111,112} This erodes the possibility for scientists to engage in long periods of hands-on research – a feature highlighted by Bell Labs researcher and Turing award Richard Hamming in *You and Your Research* as essential for producing the highest quality work.¹¹³

While many researchers excel in auxiliary roles, it should not be a near-universal requirement for our top scientific minds to devote chunks of time to lecturing, managerial tasks and the constant pursuit of continued funding.

As observed by Thomas Fink, founding director of LIMS, Great Britain would not win many medals if elite athletes were required to spend a majority of their time and energy coaching and pursuing their own sponsorship deals in the prime of their careers.¹¹⁴ Top researchers should expect no less focus and support, which requires environments where they can be fully immersed in research. This principle is embedded in the structure of LIMS. It was born out of a small group of theorists' desire to spend more time doing science, and emphasises enabling peak performance, full-time research (no formal teaching takes place) and first-rate support, with dedicated operations staff handling most of the non-scientific work.

Full Internal Funding and Internal Review

The ability to shield researchers from external pressures was made possible by institutional design choices – most importantly stable, internally allocated core funding. The vast majority of funding was conferred directly on the organisation to distribute as it saw fit. There were essentially no applications for external grants, allowing a culture of independence and free thinking to flourish. This is still the case for the LMB, where core funding is granted by the MRC for a period of five years, and distributed between labs and projects by internal processes.

Long-term, flexible core funding empowers the local scientific leadership to deliver their aims according to “coal-face” experience, removing reliance on repeated grants and precarious short term funding cycles. It is due to such a

funding arrangement that the Francis Crick Institute “can make decisions on the ground that are important for research and we are not constantly referring up several layers elsewhere”.

Closely linked to core funding is the principle of internal review. Researchers at these institutes were evaluated periodically by internal colleagues who knew their work in depth, rather than distant committees and review boards. Writing in *Nature* in 2012, Min-Liang Wong highlighted: “Four of the greatest discoveries of the 20th century – the structure of the atom, quantum mechanics, the theory of relativity and the structure of DNA – were made without project reviewers or grant-giving agencies.”¹¹⁵ To that list we could now add some of the most important breakthroughs of recent decades: AlphaFold, the deep reinforcement learning underlying AlphaGo and the transformer technology underlying large language models.

The combination of core funding and internal review facilitates disruptive work for the following reasons.

- **High level of flexibility.** Though core funding is often not entirely unconstrained, its allocation is highly flexible, allowing institutes to respond dynamically to opportunities. Central bureaucracies may be good for scale, but they are not optimised for speed, flexibility or exploiting the kind of serendipity that can spark breakthrough discoveries.
- **Long-term, high-risk research.** Core funding removes the pressure for researchers to produce steady, quantifiable short-term results aligned with short-term impact to secure ongoing grants and keep projects afloat, freeing them from reliance on central bureaucracies – which are rarely renowned for risk appetite or flexibility. Financial stability enables researchers to pursue more ambitious, long-term endeavours.
- **Less constrained research agendas.** Internal funding allocation removes the incentive to reactively pursue fashionable research areas or consensus-driven paradigms prioritised by research councils. Disruptive ideas are, by definition, counter to prevailing scientific opinion of worth and/or feasibility, rendering them challenging to fund through traditional mechanisms that rely on review by peers.

Researchers consistently report that current grant-based funding structures limit their scope and ambition. In the Wellcome Trust study, 75 per cent of UK researchers said creativity was stifled by a funder's emphasis on impact. A survey conducted in the wake of the US-based Fast Grants project found that 78 per cent of respondents would change their research programme "a lot" if they received the same level of funding but could spend it in an unconstrained manner.¹¹⁶

This points to a deeper structural problem: the tendency for consensus-orientated funding decisions to direct the evolution of scientific inquiry. Grant-based funding structures can encourage conformity, particularly in hypercompetitive fields, where proposals must appeal to peer reviewers whose judgements may reflect entrenched or conservative views of what is feasible or worthwhile. In many cases, a grant cannot be resubmitted once rejected, and so researchers are strongly incentivised to write grants that will appeal to their peers.¹¹⁷

These issues were predicted in 1948, prior to peer review becoming widespread, when physicist Leo Szilard wrote a satirical short story, *The Mark Gable Foundation*.¹¹⁸ In the story, an individual concerned by the dangers of scientific progress seeks to slow science down by increasing science funding, but introducing committee-based peer review for its allocation. He wrote: "First of all, the best scientists would be removed from their laboratories and kept busy on committees passing on applications for funds. Secondly, the scientific workers in need of funds will concentrate on problems which are considered promising and are pretty certain to lead to publishable results. For a few years there may be a great increase in scientific output; but by going after the obvious, pretty soon Science will dry out. Science will become something like a parlour game. Some things will be considered interesting, others will not. There will be fashions. Those who follow the fashion will get grants. Those who don't, will not, and pretty soon they will learn to follow the fashion too."

“The thing is to have no discipline at all. Biology got its main success by the importation of physicists that came into the field not knowing any biology”

Some of Szilard’s fears have proven to be justified. Conventional funding structures tend to favour researchers who continue projects within fields in which they have already produced “high-impact” results and established a reputation based on favourable metrics.¹¹⁹ This discourages them from venturing into unfamiliar areas where their perspectives, knowledge and technical expertise could stimulate conceptual breakthroughs. Such cross-pollination has historically resulted in leaps forward – Brenner epitomised an interdisciplinary approach in his own career, arguing that: “The thing is to have no discipline at all. Biology got its main success by the importation of physicists that came into the field not knowing any biology and I think today that’s very important. [...] I always work in fields of which I’m totally ignorant.”¹²⁰

For the benefits of core funding to be fully realised, it must be sufficient to completely remove dependence on external grants. Institutes that have partial support but remain tethered to traditional grant systems remain vulnerable to the pressures predicted by Szilard, which can undermine disruptive research. Labs with full internal funding and autonomy can create the necessary space for researchers to go against the grain, resist premature justification, and freely roam between disciplinary boundaries.

No Pressure to Publish in Scientific Journals

Scientists dependent on grant funding face ongoing pressure to publish in journals. Centralised funding bodies’ reliance on metrics such as citations and impact factors is necessary because distant bureaucracies are too far

removed to obtain more than a fragmented, low-resolution picture of the research environments they seek to assess. The pressure to maximise research metrics has created a culture known as “publish or perish”, which is largely incompatible with the type of research we seek to foster for the following reasons.

- **Constrained ambition of research agendas.** Scientists are not incentivised to pursue risky avenues of research when their career progression depends on the steady publication of papers, which generally only publish positive findings. In 2013, Nobel laureate Peter Higgs – who published fewer than ten research papers after his visionary work in 1964 predicting the existence of the Higgs boson particle – expressed doubts that similar breakthroughs were possible in the academic climate of today, primarily because of the need for scientists to continuously churn out papers.¹²¹ Brenner said something similar of his LMB colleague, double Nobel laureate Sanger: “A Fred Sanger would not survive today’s world of science. With continuous reporting and appraisals, some committee would note that he published little of import between insulin in 1952 and his first paper on RNA sequencing in 1967, with another long gap until DNA sequencing in 1977.”
- **Citation-chasing shapes the direction of scientific knowledge.** Reliance on citations as a metric for value creates a strong gravitational pull towards popular fields and trendy topics, which are likely to accumulate more citations in the short term.^{122,123} Such popular fields are not necessarily the most robust or important; this bandwagon effect can happen even in fields with shaky foundations. Research that charts completely new territory might not accrue many citations – certainly not on the short timescales required for promotions or grants.¹²⁴ This results in major areas of potential progress being neglected.

Publication and funding pressures contribute to a critical systemic problem: modern research is substantially non-replicable.^{125,126} A high proportion of published research findings are not reproducible^{127,128,129,130,131} – a situation fuelled in part by a system wherein productivity is rewarded over accurate scholarship.¹³² A recent survey found that three-quarters of biomedical

researchers believe science faces a reproducibility crisis, with “pressure to publish” cited as the leading cause.¹³³ Similarly, the Wellcome Trust report noted that UK researchers displayed “a prevalent opinion that funding criteria were a core driver of research misconduct, such as the embellishment or distortion of data”.¹³⁴ As Nobel laureate Randy Schekman observed when announcing his intention to boycott “luxury” scientific journals, pressure to publish in such journals “can encourage the cutting of corners and contribute to the escalating number of papers that are retracted as flawed or fraudulent”.¹³⁵

Such an incentive system is the antithesis of one that supports projects with a high risk of failure – where failure is defined as a lack of papers or citations in the short term. If taking risks to pursue unusual ideas creates a high chance of failing to publish, and thus your career stalling, in many cases risks will not be taken. The core-funding/internal-review model liberates researchers from publishing pressures, allowing them to report the results of their investigations in a manner that does not shape or disrupt the form of those investigations.

Minimal Bureaucracy

By trusting institutions to manage their own budgets and evaluate researchers internally, much of the administrative overhead now baked into competitive grant systems was eliminated.

Today, researchers spend more than 40 per cent of their time managing administrative tasks related to grants.¹³⁶ With overall success rates across major UK funders now hovering around 20 per cent – and often far lower for individual schemes – most of that time is spent preparing unsuccessful applications.¹³⁷ This inefficiency frustrates researchers and slows everything down. Centralised funders frequently struggle to convene reviewers with relevant expertise¹³⁸ and application-processing times can stretch for many months. On average it takes 170 days for the MRC and 216 days for the Biotechnology and Biological Sciences Research Council to approve a funding application.¹³⁹

But grant applications are not the sole contributors to this burden. In the 2022 Tickell review of research bureaucracy, institutions' own processes were the most cited source of excessive red tape, with assurance requirements from funders and regulators also highlighted as major contributors. Per the review, "unnecessary bureaucracy exists in every part of the UK research system", and the cumulative burden has escalated steadily.¹⁴⁰ What was once modest – in the 1970s, the LMB submitted a streamlined five-page report to the MRC every five years – has ballooned into what the review terms a "just in case" culture: multiple, overlapping and excessive demands for documentation. Today, the LMB must produce five 500-page reports at each evaluation cycle.

Bureaucracy is often cited by scientists as a deterrent to pursuing an academic career.¹⁴¹ According to *Nature's* 2021 Career Satisfaction Survey, 41 per cent of mid-career researchers reported that "organisational politics or bureaucracy frequently or always frustrated their efforts to do a good job". Minimising bureaucracy isn't just a funding issue – it's a question of design, and should be addressed explicitly in the governance of any institution built for disruptive research.

Why These Principles Are Major Talent Magnets

The transformative institutes of the 20th century were strong magnets for elite and diverse global talent. In part this was due to their distinctive research visions, but it was also due to their ethos. This remains true according to Hugh Pelham, former director of the LMB, who had this to say in 2013: "Senior scientists are ... attracted by the LMB ethos. For example, Venki Ramakrishnan, who shared the Nobel Prize in Chemistry in 2009 for his work on the ribosome, moved from the US to the LMB because it offered the long-term support and relative security that he felt was needed to tackle a problem as ambitious as determining the structure of the ribosome."¹⁴²

"All we have to do is create opportunities for

those who want to take risk. If we start funding this, there will be a long line of young people who are willing to participate, and it will release a huge energy which has been so far suppressed”

Garry Kasparov, 13th world chess champion, said the following to one of our authors in an interview that informed this proposal. “All we have to do is create opportunities for those who want to take risk. If we start funding this, there will be a long line of young people who are willing to participate, and it will release a huge energy which has been so far suppressed. That’s why I’m trying to promote this message.”

The need for new organisational structures presents a huge opportunity. New structures would open up diverse roles, enabling a greater number of talented people to find work to suit their skill set. At the same time, recent policy shifts in the US have created a window for the UK to position itself as a top destination for international research talent. To capitalise on this, the UK must offer something meaningfully different to its competitors and maximally appealing to researchers.

The features outlined above allowed distinctive internal cultures to emerge, as labs had substantial freedom to do things their own way. We now consider how the model labs organised their internal communities.

3. A COMMUNITY APPROACH

In modern academia – particularly in leading institutions – the PI system creates large, hierarchical structures with significant power concentrated in the hands of a few senior researchers. But that has not always been the case.

Small Teams and Flattened Hierarchies

Researchers at our model institutes worked in small, self-organising teams that operated more like miniature startups than traditional academic units. Everyone, including senior scientists, worked at the bench. Projects were primarily initiated from the bottom up via spontaneous self-organisation, rather than from top-down mandates, and group sizes were actively restricted. The institutes operated with minimal formal organisational structure and adopted flattened hierarchies, distributing power and decision-making among team members instead of concentrating it at the top.

Today a department at the Massachusetts Institute of Technology or Harvard will contain labs with 50 to 100 trainees working under a single PI, while labs at Oxford and Cambridge have 30 to 60. This is not yet the norm, but the steady increase in lab group sizes is one of the most universal trends

in science and technology over the past 50 years^{143,144} – a trend that is increasingly recognised as damaging to the research system, in part because it contributes to career structures becoming ever more sharply pyramidal.¹⁴⁵ These large labs often have a competitive advantage as they can gather resources and equipment. However, as the model labs demonstrate, such advantages can be matched – or exceeded – through shared infrastructure and collaborative design.

FIGURE 3

The 20th-century’s standout labs and their small staff counts

Institute	Group sizes
Bell Labs	Everyone, including legendary figures like mathematician Claude Shannon, was titled “member of the technical staff”. Members of technical staff had a technical assistant, making a group size of two or sometimes three. In biology the groups were very slightly larger. Nobel laureate astrophysicist Arno Penzias remarked, “There were no walls and everybody talked to everybody.”
Xerox PARC	There were no senior researchers, making the group size effectively one person. The environment was managed by Robert Taylor, who kept the group harmoniously productive.
LMB	There was the notion of a lab head but it was very informal; they largely served as recruiters for what would now be called junior fellows. Labs had a maximum of six researchers, often fewer. Fred Sanger did the research for his second Nobel Prize with his own hands, while fellow Nobel laureate Max Perutz continued doing his own hands-on research well into his 80s.

Eric Betzig, a Nobel laureate from Bell Labs who pioneered super-resolution microscopy, said, “Incentives are everything, and the incentives and small group ethos at Bell maximised creative output.”¹⁴⁶

The effects of smaller teams and flattened hierarchies on innovative capacity can be summarised thus:

- **Small lab groups promote disruptive work.** A 2019 article in *Nature* which analyses more than 65 million papers, patents and software products demonstrates that “smaller teams have tended to disrupt science and technology with new ideas and opportunities, whereas larger teams have tended to develop existing ones”. The article also notes that

small teams generate more ideas, tend to be more tolerant of risk and that “Nobel-Prize-winning articles significantly oversample small disruptive teams”.

- **Flat hierarchies and self-organisation expand the idea space.** Likewise, flat, egalitarian teams have been found to produce greater novelty, whereas tall hierarchies are less disruptive and more likely to develop existing ideas. Tall hierarchies also increase productivity for those at the top at the expense of those beneath.

The Hierarchy of the Current Model Disenfranchises Young and Junior Scientists

The innovation of the post-war institutes was in part fuelled by the empowerment of the young. At the early LMB and Bell Labs, young researchers were free to pursue their own ideas and were expected to take responsibility for their work. The average age of Bell Labs members of technical staff barely shifted from 37 years of age.¹⁴⁷ Reflecting on Xerox PARC, Kay told some of the authors of this paper, “I was the oldest researcher at Xerox PARC – I was 30.”

“The young have a great advantage in that they are ignorant. Because I think ignorance in science is very important”

Inexperience can be a virtue. While academia is predicated on the value of knowledge, the benefit of ignorance is underrated when it comes to disruptive work. As Brenner wrote, “I strongly believe that the only way to encourage innovation is to give it to the young. The young have a great advantage in that they are ignorant. Because I think ignorance in science is very important. If you’re like me and you know too much you can’t try new

things ... the most important thing today is for young people to take responsibility, to actually know how to formulate an idea and how to work on it. Not to buy into the so-called apprenticeship.”¹⁴⁸

In the collective experience of some of the authors of this paper in elite British universities, many exceptional early-career scientists are leaving research due to disenfranchisement. In the 1960s it took 35 years for 50 per cent of those entering science to leave the field; in the 2010s it took only five years for the same percentage to move on.^{149,150} Commonly cited reasons include low pay, difficulty working on their highest priority research, insecure and highly competitive career dynamics (only 3.5 per cent of UK PhD students secure a permanent research position at a university)¹⁵¹ and disillusionment with a system that contains no mechanism for recognising exemplary, inventive, reproducible science that yields null or negative results, which creates perverse incentives.

Additionally, in many fields, junior scientists devote much time to repetitive tasks or routine maintenance that may be better allocated to technical-support staff, automated systems or contract research labs. The protracted, sometimes two-decade-long “training” periods that characterise academia, coupled with the misemployment of junior scientists as cheap de-facto technicians (a situation termed “the slavery of graduate students” by Brenner¹⁵²) squanders their skills and discourages them from pursuing research careers.

Scientists in their 20s and 30s have made many of the greatest scientific and technological advances in history. Albert Einstein, Marie Curie, Ada Lovelace and Isaac Newton each made groundbreaking discoveries by their early thirties. Watson, Crick and Rosalind Franklin were 25, 37 and 31 respectively when they conducted the research that would unravel the double helical structure of DNA. The average age of Manhattan Project scientists was 25,¹⁵³ the flight controllers for the Apollo 11 mission averaged 28,¹⁵⁴ and the first eight ARPA directors had an average age of 42.

In the UK today, fewer than 1 per cent of applications for UKRI funding come from researchers under 30. They receive less money where funding is granted, and the median award amount increases with age (peaking with applicants aged 60 plus).¹⁵⁵ The average age of admission for a new fellow joining the Royal Society is 61, up from 44 in 1940;¹⁵⁶ just 14 per cent of Royal Society fellows are under 60.¹⁵⁷ The average age of a new fellow joining today is about 50 per cent higher than the average age of the Royal Society's founders.

This problem is global, highlighting the scale of opportunity. In 1980, fewer than 1 per cent of PIs funded by the National Institutes of Health (NIH) were over 65. By 2014 this had risen to almost 10 per cent, while the proportion of PIs under 36 fell from about 21 per cent to 2 per cent.^{158,159} Considering the past half century of research in the biomedical sciences, the average age at which previous Nobel laureates made their prize-winning discoveries was lower than the current average age a researcher receives their first grant from the NIH.¹⁶⁰

It is widely accepted that entrepreneurs in their 20s can found and run multibillion-pound tech companies, yet it is almost unheard of to find a PI under 30 with UKRI backing. The founders of Apple, Facebook, OpenAI, Google and Microsoft were of an age when people would still be in early-career academic “training”. Recently, major Silicon Valley US labs such as Arcadia, Arc, and FutureHouse – together collectively backed by well over a billion dollars – were founded by people at a similar age to a UK junior professor or younger. Yet public funding bodies generally consider the stakes too high to invest in scientists without a track record obtained through a PhD, multiple postdocs and perhaps a fellowship.

“In times of change, learners inherit the Earth,
while the learned find themselves beautifully

equipped to deal with a world that no longer exists”

In fields with very clear, stable knowledge, such as classical mechanics, seniority can be very useful, especially for mentoring. But in fields that are rapidly evolving beyond recognition, or in fields where progress has stagnated, that knowledge can leave you trapped with the wrong “expertise”.¹⁶¹ As US philosopher Eric Hoffer remarked, “In times of change, learners inherit the Earth, while the learned find themselves beautifully equipped to deal with a world that no longer exists.” The AI revolution, almost entirely driven by people in the first half of their careers, is a strong case in point.

The current arrangement – large, senior-led labs where credit generally flows upwards – is not the only model for disruptive science, nor is it the optimal one. If more field-creating work is desired, then small, autonomous teams should receive backing, supported by shared infrastructure – as was the case in the 20th-century exemplar labs. The UK should employ this model to convert the disenfranchisement of exceptional early-career scientists into a national advantage.

Innovation Is Bred in Social Research Communities With Strong Core Support

A defining feature of the model institutes was prosocial and efficient resource allocation; the phrase “egalitarian meritocracy”, used to describe Bell Labs, captures this well.¹⁶² The following design choices had powerful effects on research culture.

- **There was strong support from technical staff.** Each institute had cutting-edge core facilities, accessible to all labs; there was no need for labs to individually expand in order to acquire funding to purchase equipment or gain access to technologies. This setup required highly

valued core facility staff members providing technical know-how and continuity, and created a “tradition of freely offered help, advice, reagents, ideas facilities”.¹⁶³ A greater emphasis on world-class technicians and abundant support staff meant a rich mix of toolmakers, scientists and engineers working alongside one another.

- **Internal hiring was welcomed.** Bell Labs and Xerox PARC expanded primarily via internal promotion, encouraging a culture where mentoring was critical to collective success. This culture of promotion from within helped to consolidate each institute’s unique character and processes.
- **Institutes were social organisms.** Bell Labs, Xerox PARC and the early LMB were not simply workplaces – they were rich social communities. Horizontal collaboration and informal intellectual exchange were embedded into the fabric of daily operations.

The dominant academic model differs from this approach in several ways.

- **The system blocks horizontal collaboration.** In the academic ecosystem, PIs are (to a degree) in competition with one another. This means there is no real mechanism for horizontal collaboration between PhD students and postdocs in different labs, other than with the permission of their PI and within defined parameters. There is limited scope for a group of junior researchers to unite in a startup-like manner to pursue a research agenda at odds with the wishes of their PIs.
- **Each lab accumulates its own equipment, reagents and material.** Many academic departments benefit from strong core facilities, but these are often confined to equipment that is either used infrequently by individual labs, too expensive for even large labs to own themselves or highly technical requiring the aid of a technician. Smaller pieces of equipment, reagents, materials, biological samples and so on, are accumulated by individual labs and are often ring-fenced for that lab, even if rarely in use. This is duplicated in neighbouring labs, leading to huge quantities of wasted resources and a system that incentivises empire building.

As organisations naturally drift towards prestige-based, top-down structures, creative environments require continual, conscious correction and renewal. Maintaining flattened hierarchies is not just about reducing bureaucracy, but also structurally embedding trust, autonomy and intellectual freedom within an organisation.

The first director of the LMB, Perutz, wrote, “Every now and then I receive visits from earnest men and women armed with questionnaires and tape recorders who want to find out what made the Laboratory of Molecular Biology in Cambridge so remarkably creative ... creativity in science, as in the arts, cannot be organized. It arises spontaneously from individual talent. Well-run laboratories can foster it, but hierarchical organization, inflexible, bureaucratic rules, and mounds of futile paperwork can kill it.”

No institutional model can reliably churn out highly disruptive research, or guarantee that a lab will produce field-generating work. But the elements outlined above would, we believe, give the UK the best chance.

Funding Designed for Disruption

How was it possible for these institutions to operate in this distinctive way? Partly because they were products of a different time, before the global trends of rising bureaucratisation, disempowerment of junior scientists, “publish or perish” and more arose. But this is not the full story, and nor does it explain why modern efforts to resist these trends struggle to recreate the kind of internal culture that those institutions sustained.

We believe a core reason is that each of our exemplar labs received their core funding from a single source. Xerox PARC was supported by Xerox’s monopoly in photocopying, Bell Labs by the AT&T communications monopoly and the LMB by the MRC. It is likely that this was influential in enabling their successes for the following reasons.

- **Long-term funding security.** With a guaranteed and sustained revenue stream, institutes could greenlight projects that might take decades to bear fruit, unconstrained by short-term market pressures or demands for immediate returns.
- **Reduced friction and bureaucracy.** Funding from a single provider streamlined decision-making and administration.
- **Clarity of vision.** Single-source support prevented the institutes from being pulled in different directions by funders with competing interests or ideas.
- **Protection from external interference.** Institutes avoided interference in research programmes by investors seeking returns, preserving the integrity and autonomy of their research programmes.

The more sources of external funding that an institute relies upon, the more likely it is to be pulled back to the mainstream culture, which reflects the dominant funding sources. In the 20th century, monopolies such as AT&T and Xerox could afford to operate in a functionally altruistic manner, and not fully capture the value of their many contributions, or of the enormous societal benefits that their research paved the way for. While this could be called “private funding”, it was markedly different from the funding provided in a competitive market.

With the downfall of these types of monopolies,^{164,165,166} this particular type of research organism was largely lost – though it is worth noting that the UK’s prized research asset, DeepMind (the founders of which studied Bell Labs as an inspiration), is owned and funded by Google, with other pioneering AI labs similarly supported by major tech giants.

Is the new norm that only US tech giants – insulated from short-term profit pressures – can support this kind of research? Competitive markets struggle to sustain such activities: the timelines are too long, the risks too high, the returns too diffuse and the incentives too tightly coupled to near-term financial returns. In recent decades, industrial R&D has increasingly shifted away from high-risk, breakthrough science towards short-term,

market-driven projects. For example, the percentage of R&D 100 awards going to Fortune 500 companies decreased from 47 per cent in 1975 to just 6 per cent in 2006.

The decline of the industrial research lab over the past 30 years means it is especially important that the state takes up the mantle of funding disruptive invention. Indeed, in many ways the government is much better placed than the private sector to support this sort of long-term investment in humanity's future through science and technology – at least in an anchoring role, as it did with programmes such as Apollo. For a broader discussion of the decline of the industrial R&D lab, see the essay “The Rise and Fall of the Industrial R&D Lab” by Ben Southwood.¹⁶⁷

Having covered the core features shared between the 20th-century model labs, we now turn to contemporary efforts to revive some of these features.

Lessons From Contemporary Independent Institutes

Recent years have seen the creation of several independent research organisations explicitly founded to address shortcomings in traditional models. Several non-profit funding agencies have supported innovative institutions that have become international leaders in their field, including the Champalimaud Centre for the Unknown, the Sainsbury Wellcome Centre, Janelia Research Campus, The Francis Crick Institute and the CZ Biohub Network. The Whittle Laboratory at the University of Cambridge has recently established the Bennett Innovation Lab – an independent institute within the Whittle Lab designed to address key shortcomings of the academic system. Many of these institutes benefit from excellent core facilities and top researchers, who are increasingly moving over from academia.

However, despite these strengths, many of these institutes have not been able to fully develop their own culture. This limitation stems largely from their continued reliance – both structurally and culturally – on academic norms. Ultimately, many researchers have no choice but to return to the existing academic system. This manifests in several ways.

- **Hierarchical laboratory structures.** These independent institutes have frequently imported traditional academic structures without fundamentally rethinking incentive structures or how to organise their research, thus retaining large, fixed lab groups and hierarchical systems.
- **Dependence on traditional academic careers.** Many researchers, particularly postdocs, anticipate a return to traditional academia. Consequently, they know they will be judged by external measures of success – most obviously publication – which leads to the institutes, and the researchers within it, competing based on metrics and standards set by the external culture.
- **Reliance on external grant funding.** Relatedly, with exceptions, these institutes are still dependent on external grant funding, which in turn influences the setting of short-term objectives and the avoidance of high-risk/high-gain projects.
- **Limited internal promotion.** Members are rarely promoted from within, which can prevent the growth of a cohesive culture. The only constant is senior leadership, meaning junior people lack a stake in the collective's success. This feeds into the issue outlined above: if researchers know they will have to return to the dominant incentive system, they will begin to orient towards it.
- **Absence of a cohesive vision.** These institutes rarely have a vision of the kind seen at the LMB, PARC and Bell Labs, instead typically choosing a very broad portfolio or working in established disciplines.

These are not critiques, per se; rather, we seek to highlight how these new ventures differ from the model labs explored above. However, some projects have taken more decisive steps away from traditional academic structures, with notable examples including Future House, LIMS, Arcadia and the Arc institute.^{168,169,170,171}

The non-profit Arc Institute was founded in 2021 by biochemist Silvana Konermann, bioengineer Patrick Hsu and tech entrepreneur Patrick Collison, who also collaborated on the Covid-19 Fast Grants initiative. Arc is a biomedical research institute that aims to overcome the limitations of conventional research organisations by pursuing ambitious, long-term

scientific projects focused on complex human diseases. It has emerged as a leading example of the applied metascience movement,¹⁷² demonstrating how new organisational models can revitalise research ecosystems.

Arc's operating model is underpinned by substantial philanthropic backing, having secured an initial commitment exceeding \$650 million. This significant guaranteed funding allow it to provide eight years of renewable, no-strings-attached full funding to its Core Investigators, along with state-of-the-art core technologies such as AI-driven drug-discovery platforms, advanced sequencing and computational tools.^{173,174} Its approach has already reaped rewards, with early successes including a novel gene-editing method for massive DNA rearrangements using "bridge recombinases",¹⁷⁵ the creation of AI-designed antibody molecules¹⁷⁶ and Evo 2 – an AI that can "model and design the genetic code for all domains of life",¹⁷⁷ allowing the generation of lab-viable viral genomes.¹⁷⁸

Arc's rapid operational growth, ability to attract exceptional talent and early scientific breakthroughs underscore the potential of intentionally redesigned research environments. Other world-leading labs have likewise demonstrated significant success. Each of the scientists who shared the 2024 Nobel Prize in Chemistry – Demis Hassabis and John Jumper, the CEO and Director of DeepMind,¹⁷⁹ and David Baker, who founded the Institute for Protein Design^{180,181} – run research institutes explicitly inspired by Bell Labs' culture and operating principles.

For a new generation of disruptive labs to succeed, conscious steps must be taken to ensure they can develop their own momentum and identity. This requires mission-oriented internal design, a governance model that prevents drift back into the academic status quo and sufficient cultural gravity to attract top global talent.

Achieving this will require a new kind of research entity, built for this purpose from the ground up. It is this project that we focus on in the next chapter.

04

Lovelace Disruptive Invention Labs

In this chapter we bring the strands of chapters 2 and 3 together to propose a new kind of research institute for the UK: a complementary model designed to sit alongside existing institutions. Rooted in historical precedent, it responds to the systemic constraints created by over-reliance on a single mode of public R&D and offers a dedicated vehicle for supporting disruptive science. These institutes would concentrate creative scientists, engineers and technicians into a network of interdisciplinary labs to work side by side on nascent windows of technoscientific opportunity.

We emphasise two caveats upfront. First, this proposal seeks to complement and strengthen universities, not replace them. Many core functions of research – and the wider scientific ecosystem – are best conducted within the university system. The success of our model depends on the existence of a strong, healthy academic sector, as well as its infrastructure, talent and long-term capacity. Conversely, universities stand to gain from proximity to our proposed disruptive labs, as demonstrated by the relationship between the LMB and the University of Cambridge. However, it is crucial that these labs are administratively and financially independent of universities.

Second, elements of this model already exist within parts of the independent research ecosystem, and similar ideas have surfaced in recent policy work. However, to our knowledge no existing UK public institution combines the full set of features we propose. We aim to build on such precedents and push further, integrating key principles into a purpose-built structure that is more deliberately and durably distinct.

Several areas would require particular attention from funders and policymakers: prioritising junior scientists; enabling novel modes of publishing and data sharing; and establishing alternative career pathways that move beyond the incentive structures of academia, while remaining rooted in early-stage, pre-commercial exploratory research.

A New Research Laboratory Network

The call for new models of scientific research is gathering momentum within the UK's R&D ecosystem and has already resulted in several notable successes.^{182,183} The creation of ARIA in 2022 was a significant step forward, marking a clear intent to diversify and strengthen the national research infrastructure. Similarly, the previous government's commitment of £50 million through the Research Ventures Catalyst fund demonstrated an understanding of the value of exploring alternative research models.^{184,185} However, while this investment amounted to an acknowledgement of the need for new entities, it was limited in scale, being roughly equivalent to one year of the LMB's budget.

We are not the first to identify that a gap exists in the UK R&D ecosystem, nor to propose institutions to fill it. There have been several recent proposals for national networks of non-academic research laboratories, spanning theory, discovery and engineering in areas of emerging technoscience. These calls come from individuals with experience working in atypical research institutions including Xerox PARC and Bell Labs. Here are some of those proposals.

- **Lovelace Laboratories:** Originally proposed in 2020 by a group of early-career researchers – convened by advisors in 10 Downing Street, including a co-author of this report – this precursor to our concept called for a network of laboratories spanning discovery and invention, offering long-term support to promising junior scientists. Inspired by institutions such as Xerox PARC and newer models such as Janelia, the proposal was centred around purpose-built environments for frontier research.
- **Disruptive Innovation Laboratories:** A proposal submitted to the Nurse review by Robert Miller and Eoin O'Sullivan from the University of Cambridge.¹⁸⁶ They call for a network of laboratories to “bring together a critical mass of highly talented people with the right skills, tools, culture and environment to work effectively at the interface of scientific discovery and engineering development. These labs would target critical early

stages in the lifecycles of disruptive technologies, where there are ‘windows of opportunity’ to translate national scientific strengths into global technological and industrial leadership.”¹⁸⁷

- **NewLabs:** A proposal by Sir Andrew Hopper, former vice-president of the Royal Society, for a network of laboratories working on novel areas of technoscientific opportunity. NewLabs would pursue breakthrough advances with a specific emphasis on deliverable technological prototypes, with design features strongly inspired by PARC. While the proposal is not publicly available, it was a significant influence on the authors of this paper.
- **Bell Labs 2.0:** Narayanamurti and Odumosu’s *Cycles of Invention and Discovery*, and Narayanamurti and Tsao’s *The Genesis of Technoscientific Revolutions* outline priorities for new research environments. Tsao has recently extended this with a proposal for a new R&D lab structure called Bell Labs 2.0.¹⁸⁸ While US-focused, its arguments extend to the UK.
- **X-Labs:** A recent proposal for a US network of X-Labs included in the “The Techno-Industrial Policy Playbook” by Institute for Progress co-founder Caleb Watney.¹⁸⁹

Common features of these proposals can be integrated with our earlier analyses to develop a concrete institutional model. We have done this in three parts, with particular emphasis on how this differs from typical approaches to science:

1. Physical laboratories focused on nascent areas of technoscientific opportunity.
2. A non-academic research fellowship.
3. Sustained support for a research community.

Together, these outline the proposed model for the network of Lovelace disruptive invention labs.

A Physical Laboratory Focused on a Nascent Area of Technoscientific Opportunity

At the core of our proposal is a network of physical laboratories, each focused on a nascent area of technoscientific opportunity.

RESEARCH LABORATORIES ORIENTED TO A VISION

Each lab would share a unique, broad, unifying research vision while retaining substantial latitude to explore promising new directions as they emerge. This approach is more expansive than that of FROs, which typically have a well-defined and time-limited mission, but narrower than institutes such as the Crick, which cover broad scientific domains such as biomedical research.

We refer to this middle ground as “vision-oriented research” – a term used by Kay of ARPA/PARC to define labs with a clear ambition to bring about meaningful change in the world through technological invention, but without a fixed blueprint for how that change will be realised, nor immediate commercial considerations.

A network of labs would be composed at the portfolio level, with the expectation that only some would lead to industry-birthing advances, but all should push the frontier of a promising technoscientific domain. While detailed selection methods are beyond the scope of this initial proposal, candidate areas would potentially meet one or more of the following criteria.

- **A recent or foreseeable breakthrough unlocking a new field.** For example, the development of optimised channel rhodopsin (light-gated ion channels) in the mid-2000s unlocked the field of precision neurotechnology and transformed the study of neural circuits.
- **A niche in global research attention.** Some promising novel research directions, while stemming from more established disciplines, remain underfunded and institutionally unsupported. This offers the potential to occupy a global niche.

- **A bottom-up network of researchers with a nascent research vision.**

Recent examples include the renewed interest in bioelectricity in the field of regenerative medicine, the mechanistic basis for the mind-body relationship, and the work of Bret Victor's group extending the ARPA-PARC vision to embodied computing.

- **A unique national advantage.** Areas in which the UK holds a structural lead – such as those enabled by the NHS or UK Biobank – could be prioritised.

- **Necessity for public investment.** A strict criterion would be a clear case explaining why such a laboratory and its research could not occur solely through private, for-profit capital.

During earlier scoping exercises that some of our authors conducted in 2020, areas such as cellular learning and technologies for interoception were example topics, which have since grown greatly in prominence.¹⁹⁰

Some of these topics are now funded by ARIA, suggesting that it is possible to be ahead of the curve when it comes to global interest.

EMPOWERED LEADERSHIP WITH THE FREEDOM TO EXPLORE

To realise the potential of vision-oriented research that can defy consensus, laboratories must be entrusted with a high degree of autonomy, which means embracing empowered leadership models.

The lab would be headed by empowered directors, who would have broad and significant discretion over team composition, resource allocation and research direction, with the ability to make key decisions independently of the parent organisation. This is a sharp departure from the fragmented, slow-moving structures typical of conventional institutions, where decision-making is often delayed by multiple layers of approval, removed from the realities of day-to-day research.

This would mean freeing laboratories from the burden of prospectively justifying their activities to centralised funders or conforming to rigid strategies devised far from the research frontier. Instead, accountability

should be exercised through retrospective internal-review processes that evaluate the quality of decisions, rather than adherence to pre-approved plans. Progress, not paperwork, should be assessed.

Directors would be expected to make consequential resource decisions not only on how to allocate funding internally, but also when to terminate underperforming workstreams or pivot the lab's overall direction in the light of new findings or as the field evolves. The governance of Lovelace labs should be designed to support responsiveness, so that they can continuously adapt their internal structures and scientific priorities as part of the research process itself.

This requires leadership that combines scientific imagination with operational authority, underpinned by clear accountability. Directors should be chosen for their alignment with the purpose and culture of these laboratories, not simply promoted from traditional academic leadership roles.

Here, leadership is not synonymous with seniority. Many of the past decade's most transformative research environments – from OpenAI and DeepMind to Arc and Arcadia – have been founded or led by individuals much earlier in their careers than typical heads of UK institutions, often without formal managerial credentials. In the context of building something decisively different, conventional experience could actually be a constraint. The leadership of Arc, Arcadia and Future House are all in their 30s and 40s.

This role should be seen as more of a stewardship for the community, not as a vehicle for personal scientific prestige. The role is not to oversee a single research programme or build an academic empire, but to cultivate a creative, generative environment – one in which researchers have the time and freedom to think deeply, collaborate widely and explore the unknown. Gerry Rubin, describing his founding leadership at Janelia, likened the role to that of “the concierge at a very fine hotel”.¹⁹¹

Finally, while these labs might be physically adjacent to leading universities or research hospitals, they would have to be administratively independent. Their placement should be guided by scientific opportunity, not regional distribution or economic-development goals. Physical proximity can

promote collaboration and shared culture, but formal independence ensures that local leaders maintain genuine discretion, free from university politics or bureaucratic drag. This model aims to capture the best of both worlds: the vision, flexibility and urgency of entrepreneurial science with the depth, rigour and long-term orientation of public research. An interesting example of an independent institute housed within an established university structure is the Bennett Innovation lab – intended to combine the agility of a startup with the Whittle lab’s technical expertise and infrastructure.

Such freedom must be balanced by responsibility. Oversight processes – tailored to the unique model of these labs – should be in place, to ensure that those with significant decision-making authority are held to account. We explore this further in later sections.

A SINGLE STREAM OF SUSTAINED INSTITUTIONAL FUNDING

For laboratories to operate with long-term autonomy, they must be backed by a stable and effective funding model. As Kay wrote, “The goodness of the results correlates most strongly with the goodness of the funders.”

There is growing recognition that the UK’s default approach to funding public-research institutions is suboptimal. Nurse, both in his 2023 landscape review and in recent parliamentary testimony, criticised the prevailing funding model, whereby labs depend on a patchwork of contributions from multiple institutions, each with their own politics, processes and expectations. He also highlighted the recent trend for laboratory buildings to be funded, but without the sustained core funding necessary to support the actual research.

In the UK independent institute ecosystem, this patchwork reality is evident. LIMS, for example, has to date funded its physics research almost entirely through private investors and EU and US grants, rather than through access to a dedicated stream of domestic core public funding. That this has been possible is testament to the institute but it underlines that, in the absence of sufficient domestic mechanisms for supporting non-university institutions, high-potential labs are forced to assemble precarious and unstable mosaics of foreign and private support.

Empowering leadership to make locally informed, responsive decisions requires a supportive funding arrangement. Relying on multiple fragmented sources of funding presents a challenge, as it shifts decision-making power away from the lab and towards external actors, while also increasing bureaucratic overhead. As outlined earlier, all three model 20th-century institutions benefited from a single source of funding that covered all core research costs. This is also the case for Google DeepMind and Arc.

While the precise funding model may vary across Lovelace labs, we suggest three overarching principles:

1. A single source of public funding support, sufficient to support the lab's core research activities for at least 15 years.
2. Preclusion from applying for additional public-research grants.
3. Openness to philanthropic and private partnerships, which can be more confidently cultivated when a lab benefits from the agility, focus and long-term support offered by a stable core funding arrangement.

Where appropriate, the expectation may be that labs transition towards full private funding over a 15-year period, though this would not suit all cases.

We now explore how this framework supports two other core goals: a distinctive approach to attracting and enabling scientific talent, and the potential to scale influence and enable collaborative team science.

A Talent-Centric Research Organisation

In all Lovelace labs, the culture of autonomy should extend beyond leadership. By enabling a self-organising research environment, labs could maximise the use and empowerment of talent. This would also allow for a broader and more diverse definition of talent than is typical in today's academic system.

The specific mechanisms by which this culture would be instilled would vary from lab to lab, depending on research focus, the respective directors' preferences and the lab's unique requirements. This being true, we propose

three overlapping design features that could enable early labs to embody this culture: a distinctive research fellowship, a reimagined technical-staff model and a rebalanced managerial layer.

LOVELACE FELLOWSHIP

We propose that a core feature for Lovelace labs could be a distinctive fellowship programme: the Lovelace Fellowship. These fellowships would provide long-term, elite-level resources to promising, creative and highly driven junior (not necessarily young) researchers. Fellows would be subject to renewal on long-term timescales, freeing them to take the substantial risks necessary for bold discovery and invention.

The fellowship is designed as an alternative to the traditional academic pathway, whereby early-career researchers face precarity, dependence on senior sponsorship and limited independent access to resources. In conventional systems, stability and the freedom to explore come only with seniority, by which point researchers are often expected to manage large teams and navigate complex institutional politics. Only senior figures can typically accrue sufficient resources to pursue some of the more technically challenging and resource-intensive experiments, while credit and opportunity often flow upwards.

As seen in our exemplar labs, an alternative model is possible: provide sustained support directly to exceptional researchers, invest in the person rather than a predefined project and remove the rigid hierarchy and demands of tenure.

The Lovelace Fellowship would incorporate the following elements.

- **Sustained support.** Lovelace Fellows would receive something in the region of ten years of support, with a formal renewal review at year five and ongoing light-touch internal reviews, such as biannual presentations.¹⁹² Although granted significant freedom and long-term backing, fellows would not hold tenure and would be precluded from applying for external grants, maintaining institutional alignment.

- **Small, focused teams.** Each fellow would be supported in hiring one to three other researchers, akin to a startup founding team. Team size would be capped and tailored to the research topic and evolving needs, with modest flexibility based on excellence. Fellows would also be able to self-organise into larger dynamic groups for collaborative efforts. In this sense, the Lovelace Fellowship can be seen as a renewable MacArthur Fellowship for small teams.¹⁹³
- **Resource support.** Fellows would have access to a dedicated research budget sufficient to draw on core technical resources, including lab technical support, procurement of infrastructure and commissioning of external research. Support should be sufficient to allow research fellows to pursue their highest priority research, similar to the focus of an early-stage startup.

This model inverts several assumptions of conventional research organisation. Rather than tying significant resources to large senior-led groups, it provides robust, sustained support to small teams led by early-career researchers. Crucially, the structure allows top research talent to remain deeply engaged in research, not management.

To attract top talent, the likes of salary and hiring terms should be adjusted to ensure that leaders are able to attract their first-choice hires for key roles. As was the case at Bell Labs, the offering may vary substantially between researchers, based on discipline, ambition and need.

The hiring criteria for Lovelace labs should not be rigid. Rather, they should rely on the taste and judgement of directors, managers (detailed below) and existing fellows, and the needs of the lab itself – for example, for a particular piece of the research landscape to be filled or for specific expertise.

Candidates from a wide range of backgrounds should be welcomed: those who are fresh from PhDs, as well as those transitioning from industry or startups, or from senior positions in academia with the wish to get back to frontline research. This agnostic approach would allow Lovelace labs to pursue the most creative and unconventional thinkers across domains. Several broad principles would guide recruitment.

- **Promise, not prestige.** Fellows would be chosen for their potential to produce transformative research and invention over a 15-year horizon, not for their institutional pedigree or publication record – a Royal Society for active researchers.
- **Non-traditional profiles.** The freedoms of Lovelace labs should be used to identify and nurture researchers who might not produce work in a way that conforms to academic metrics: for example, those with few, if any, papers or credentials but demonstrable originality, creativity and technical skills. Some fellows might not even focus primarily on research but contribute in other ways. That could be designing tools to integrate and communicate lab findings, hosting workshops or exploring how best to communicate the lab's work, comparable with Christopher Wren's artistic role in the early Royal Society.
- **"Misfit" hires.** Relatedly, the programme would actively welcome individuals who might have left academia some time ago, never completed a PhD, or whose career path has been non-linear. Industry has long recognised the value of such talent; public R&D has not. A good example is the hiring of Eric Betzig by Gerry Rubin, founding director of Janelia. Betzig, who had previously worked at Bell Labs, had no offers of academic employment at the time, but Rubin brought him on as group leader. Within a decade he had won the Nobel Prize.
- **Team hires.** The fellowship would support joint applications from small groups of collaborators (for example, four postdocs or PhD students) who have identified a problem requiring coordinated effort. Such work may be too early-stage for a startup and too interdisciplinary, iterative and collaborative for standard academic structures, but ideally suited to the Lovelace model.

For those familiar with conventional academic lab hierarchies, the role of a Lovelace fellow may be analogous to that of a senior postdoctoral researcher in a large lab – but with much more autonomy, resources and long-term support.

TECHNICAL EXCELLENCE SUPPORTED BY DEFAULT

Dedicated technical support should be central to the operation of each lab. This has become increasingly important as scientific complexity has deepened: few, if any, researchers now possess the full range of skills required to design, execute and interpret a series of experiments entirely on their own – and even when they do, working in this way is often inefficient.

Vision-oriented research models support the use of shared technical infrastructure. In aligning researchers around a broadly common set of challenges, there are enough commonalities between the technical needs of each researcher and project such that the labs can utilise shared technical resource cores and technicians efficiently. The Lovelace model should also include non-fellow scientists: flexible, highly skilled staff who can move between projects as required, guided by evolving local priorities.

This model deliberately moves away from the pervasive but inefficient norm in which graduate students and postdocs serve as de facto technicians. Instead it restores a stronger tradition – more prominent in earlier eras and revived in institutions such as the Crick and Janelia – of dedicated technical specialists supporting scientific work. This not only frees researchers to focus on high-level inquiry but also improves the quality of technical execution through specialisation. A commitment to small team sizes is only possible where there is strong technical support.

MANAGERIAL ROLES

Replacing the traditional professorial “principal investigator” model with flatter, more fluid self-organising structures does not mean eliminating managerial roles. While hierarchical control should be minimal, managerial roles – including and in addition to the director role – remain essential.

The number and structure of managerial roles would vary by lab size. In smaller labs – such as those modelled on Xerox PARC, with about 30 researchers – a single director might suffice, with no additional management roles. Larger labs could require additional support to handle increasing complexity.

It is important to distinguish these roles from academic professorial management and more typical “directional” industrial models. Here, management is less about directing research agendas and more about enabling them, creating the conditions for others to thrive.

Managers should fulfil the following core functions.

- **Resource allocation.** Making decisions about the internal distribution of funding, space and equipment – particularly between fellows and teams.
- **Fellowship oversight.** Participating in the selection, renewal and potential transition of Lovelace fellows.
- **Infrastructure strategy.** Guiding investment in shared equipment and determining technical support priorities based on emerging needs.
- **Community stewardship.** Supporting the social and intellectual health of the lab: mentoring, resolving tensions and helping to build a collaborative culture.

These managerial roles would require a degree of technical competence, not just generalist skill. This does not necessarily mean deep specialisation in a narrow area, but rather a meaningful familiarity with the scientific domain, sufficient to exercise sound judgement on hiring, infrastructure investment and research direction. Bob Taylor’s leadership at Xerox PARC is a good example: while not a technical contributor himself, he had deep contextual understanding and exceptional judgement in shaping a high-performing team.

Equally important is what managerial roles should *not* entail.

- **They need not fall to the most senior or distinguished researchers in the lab.** Claude Shannon, despite inventing information theory, remained a “member of technical staff” at Bell Labs without taking on managerial duties. This allowed him to focus fully on his research.
- **They should not be permanent appointments.** Unlike tenured professorships, managerial roles should rotate. This allows for fresh leadership and avoids the entrenchment of hierarchy.

- **They are not roles for directing junior researchers in detail.** Project-level guidance should come from fellows or project scientists (see below). Managers exist to support the research environment, not dictate it.
- **They are not vehicles for personal credit.** In the best historical cases – Watson and Crick, Shannon, Engelbart – the principal credit for breakthrough work went to those who did the work, not their supervisors. Modern academic systems have tended towards the increased channeling of credit to lab heads or institutional leads, including in recent Nobel Prizes.

Scaling a Team-Science Approach

Lovelace labs would empower small self-organising teams, but such teams can struggle to scale their work, explore large potential solution spaces for problems requiring substantial labour and execute complex, multi-component experiments.

If Lovelace labs are to have wider impact, they would have to adopt new models for sharing their research and for evaluation – particularly when their outputs don't fit neatly within existing academic norms.

Here are three possible mechanisms designed to address these needs.

1. **Coordinated research programmes** that enable research projects of a scale and complexity not possible in small groups.
2. **Open sharing platforms** for convening and sharing of research, built for modern, AI-native modes of dissemination and exchange.
3. **Tailored laboratory reviews** that assess labs holistically through mechanisms that consider the totality of their contributions, which would be evaluated by those best placed to do so.

COORDINATED RESEARCH PROGRAMMES

Coordinated research programmes (CRPs) are envisioned as larger-scale projects that would be too substantial to be pursued by an individual fellow or through ad hoc collaboration between small groups. These projects would require dedicated budgets, more defined objectives and specific managerial control, with team sizes exceeding those of a typical Lovelace group. They are conceptually aligned with models such as Janelia's Project Teams and the Allen Institute's team-based research efforts.

CRPs are intended to fill key gaps – both within Lovelace labs and the broader research ecosystem. While small groups are well suited to generating disruptive ideas, they are less effective at scaling promising directions into fully realised research programmes. Plus, there is a well-documented global funding gap for projects in the medium-size range: those that fall between the scale of a single lab and that of major scientific endeavours such as CERN and the Human Genome Project. This “missing middle” has recently begun to receive attention through models such as FROs, as well as proposals from the likes of X-Labs in the *Techno-Industrial Policy Playbook*, produced in May 2025.¹⁹⁴

Examples of such projects already exist, typically emerging from atypical research organisations outside the mainstream academic structures. These include:

- **GCaMP brain-sensor protein series.** The GENIE Project Team at Janelia produced a suite of neural-activity sensors, the most notable of which – GCaMP – built on earlier university research to enable real-time recording of activity across tens of thousands of neurons in living animals.
- **Allen Brain Atlas.** A series of large-scale gene expression and anatomical connectivity-mapping projects from the Allen Institute for Brain Science, providing the most comprehensive maps of the rodent brain to date across multiple modalities.

- **Evo 2 Foundation Model.** Developed in 2025 at the Arc Institute, this generative-AI model enables accurate prediction of genetic variant effects and the design of novel genomic sequences, accelerating progress in biological discovery, precision medicine and synthetic biology.

Adapting this model for Lovelace labs, each CRP would be led by a fellow nominated as programme manager and supported by a group of advisory fellows. Following an application – ideally initiated by fellows, though proposals from external collaborators could also be considered – programme approval would sit with the lab director. The scale and structure of these medium-sized programmes would vary by lab, but collectively they could account for up to 25 per cent of a lab’s overall budget, potentially co-funded in partnership with other funders.

OPEN RESEARCH SHARING AND CONVENING

Research is fundamentally a social endeavour, shaped by the mechanisms through which knowledge is communicated. Yet despite its centrality, and the huge amounts of money poured into it, the infrastructure for sharing research remains largely unchanged.

Unlike in other sectors – where television, the internet and social media have transformed how information is produced and consumed – academic research continues to rely on centuries-old conventions. From the academic paper as the primary vehicle for results to the structure of undergraduate education, many aspects of scientific communication reflect inherited practices rather than optimal ones, with no actor able to change the course.

The system has, however, been the subject of considerable and growing criticism,^{195,196,197,198} which has prompted efforts to shift to new systems. Funders such as the Wellcome Trust, the Howard Hughes Medical Institute (HHMI) and the Max Planck Society united some 15 years ago to form a new journal, *eLife*, that seeks to shift norms around the accessibility of research and mechanisms of review.¹⁹⁹ Initiatives such as the open-access archive arXiv, and its life sciences equivalent bioRxiv, mean that the publication of “preprints” prior to formal research publication is now possible. Prominent

scientists such as AI pioneer Yann LeCun and leaders at HHMI have called for more substantial departures from the existing system.^{200,201} At the institutional level, Arcadia has adopted a highly novel publication approach.²⁰² We note that organisations like OpenAI and Anthropic publish their technical work almost entirely through blogs and preprints.

Given the initiatives listed above are at the single lab and funder level, a further step would be to have a laboratory network that is unified in its commitment to bringing the sharing, teaching and convening of scientific information into the 21st century.

The creation of a new network of research laboratories backed by the government would present an opportunity to reinvent how science is communicated, making it more flexible, open and diverse. This opportunity is especially powerful because the renewal and funding criteria for Lovelace fellows and projects would be established in-house, meaning researchers would not need to pursue the typical indicators of success: notably publication in leading journals, in what *eLife* co-founder and Nobel laureate Randy Schekman calls “the tyranny of the luxury journals”.²⁰³ This approach could once again position the UK as a leader in the evolution of scientific communication, as it was with the creation of the Royal Society.

Three pillars would support research sharing and convening:

1. **Open and early sharing of research and data by default.** New Lovelace disruptive invention laboratories should be expected to be proactively innovative in sharing their research results. This would include prioritising open publication venues (platforms on which the public has free access to research) and sharing data and work much earlier than is currently the norm.
2. **Educational workshops in relevant research areas.** Lovelace labs would not be required to undertake conventional undergraduate teaching but would conduct workshops and training programmes in “edge-of-the-art” knowledge, techniques and questions in their field of focus, with this constituting a core educational priority that would also benefit their research.

3. The convening of global talent in relevant spaces for research

conferences. Hosting bespoke, focused conferences and unconferences involving top researchers, as well as industry, is a powerful way to share knowledge and recruit talent.

The UK currently largely lacks these kinds of workshops, but global courses organised by labs such as the Marine Biological Laboratory in Woods Hole, Cold Spring Harbor, the Champalimaud Foundation and Janelia set a precedent. They provide valuable educational resources and put their host organisations on the map in their relevant fields, making them excellent recruitment tools. There is an argument to be made for launching labs that initially focus on running graduate teaching courses, with early fellows hosting workshops and standout attendees being recruited to populate the labs. This would be similar in approach to the early days of Xerox PARC which, through the ARPA-PARC programme, identified potential recruits through ARPA-sponsored graduate workshops.²⁰⁴

TAILORED LABORATORY REVIEW PROCESSES

Review processes are one of the most crucial levers for shaping a laboratory's culture and for maintaining trust. They set core incentives for management, researchers and funders, defining what counts as worthwhile work and over what timescale. If Lovelace labs are to scale ambitious science and encourage free, collaborative internal dynamics, evaluation processes would have to be designed to support these goals. The following principles are foundational for enabling institutional conditions in which team-based, exploratory science can thrive.

- **A portfolio mindset.** Lab performance should be reviewed in totality, not atomistically. A major breakthrough in one line of research may justify renewal, even if many other lines of research fail.
- **Expert review, not audit.** Evaluation should be conducted by first-rate researchers with direct familiarity with the kind of exploratory, high-risk work being pursued – not administrators or generalist panels. To reduce the chance of conflicts, international reviewers should be prioritised.

- **Appropriate timescale.** Review cycles should match the lab's missions. Frequent evaluation drives short-termism; longer review intervals can (perhaps counterintuitively) lead to better outcomes.

Lovelace disruptive invention labs are, at their core, a bet on the transformative potential of talented people working in the right conditions. This chapter has outlined how such conditions can be created; next comes a plan for how this vision can be realised.

The Path Forward

In this final chapter we shift focus from the proposed network of Lovelace laboratories and their design to the institutional vehicle required to bring them into being. We explain why this institutional vehicle – the Lovelace Society – would ideally take the form of a new public body, and outline the features, capabilities and powers it would require to pioneer this model for research at scale. We close by outlining a plan for the Society’s creation, detailing common failure modes that should be avoided and factors that should be considered in the society’s evaluation.

This is a sketch of an ideal path, not a rigid blueprint; political realities may require that other routes be taken.

A Society for Creating Novel Scientific Communities

Setting up and maintaining new research laboratories presents significant challenges, as evidenced by past UK efforts.

In outlining how to create a network of disruptive invention laboratories, there is an inherent trade-off between legibility and success. Planning in detail for a programme such as this would be counterproductive, detracting from its ability to operate flexibly, self-organise and ultimately find the right path for itself. However, several foundational principles and capabilities would have to be embedded in the design of Lovelace labs, and of the network, from the outset.

- **Mission-driven focus.** The institutional vehicle that creates, sustains and oversees the network of labs (the parent organisation) should have a singular, well-defined mission, ensuring all processes and hiring align with this goal.
- **Empowered frontline leadership and minimal bureaucracy.** This organisation would have to embrace a governance model that prioritises independence, speed and minimal bureaucracy. It should not require approvals from multiple layers of government, which may only be granted

once ideas have become sufficiently conventional and safe, or have even become the status quo. Instead, it should be inspired by the setup at DARPA – a \$3 billion organisation with only about 125 staff, despite having existed for more than 60 years. Its ability to remain so lean is the result of a strict culture of individual autonomy and responsibility. Like DARPA (and similar models like that of the LMB under Max Perutz), the organisation that oversees the Lovelace network should minimise decision-making by committees in favour of empowered directors and programme managers, supported by an advisory board.

- **Review, not audit.** It would have to commit to the detailed review of fellows, projects and labs through non-bureaucratic approaches. Review processes should be conducted with the ethos that reviewers being close to their subjects and their work confers resistance to getting bogged down by a reliance on crude metrics, instead allowing for a qualitative, informed and holistic evaluation. This “review, not audit” mindset puts humans – rather than paperwork – at the forefront.
- **Strategic risk-taking.** It would be critical to embrace a portfolio approach, acknowledging that a substantial portion of projects might not achieve their immediate objectives. This approach mirrors that taken by the Vaccine Taskforce. In short, one has to back some failures in order to find the big successes. Relatedly, Lovelace labs should be capable of making rapid speculative investments. Each lab should operate with a pot of unallocated funds – what Kay dubs “mad money” and Brenner called a “casino fund” – allowing for speculative investments when opportunities arise.
- **Sustained investment.** As highlighted in some of TBI’s previous New National Purpose reports, building and maintaining a high-quality and disruptive research community depends on long-term, stable investment. Researchers must be confident and secure enough to pursue their most ambitious ideas. A total budget for the parent organisation should be set for each financing evaluation period, then the responsibility for funds and resource allocation should be left to individual labs.

- **Global competitiveness and talent attraction.** The network would have to be able to compete with the best environments in the world. Institutes such as the Crick, Arc, DeepMind and Janelia (during Rubin's directorship) have succeeded by recruiting the most exceptional, internationally mobile talent. Recruiting top talent from the widest possible pool should be the focus of Lovelace's strategy and would serve as an early marker of success. To enable this, fellows' salaries should be tied as a fraction of the industrial market rate, with baseline and ceiling levels.
- **Strategic institute locations for agglomeration effects.** Lovelace labs should be in areas that maximise their potential for success: near hubs of scientific excellence, international transport links and world-class university departments. Prioritising strategic placement over regional economic or political considerations is essential to fostering agglomeration effects.
- **Optimisation for speed.** In a world defined by intense technoscientific competition, performance underwrites growth, security and influence. Supporting long-term research does not mean moving slowly: the goal should be to build physically operational labs within five years. The scale and speed of what was achieved during the second world war demonstrates how quickly such feats can be achieved.

These principles define the setup and support requirements for the organisation that oversees the Lovelace labs network. We have considered three broad approaches for its establishment.

1. AN INDEPENDENT, NON-GOVERNMENT ORGANISATION

A non-profit organisation could be created, potentially seeded by the government. This would be akin to the setup of Convergent Research, which matches philanthropic capital to FROs, and would bypass complications of government ownership.

However, this is unlikely to be the optimal approach. The UK R&D ecosystem does not benefit from the large pool of philanthropic funding that exists in the US, limiting the ability to set up an organisation of this magnitude through philanthropic means. It would be challenging to address this deficit

sustainably through government funding via an independent organisation. In considering the setup of ARIA, it was noted that once an independent organisation exists, for example as an endowed entity such as NESTA, it can be legally problematic to continually refund it without open competition. This was a core reason for the decision to legislate for ARIA, instead of using an endowment model.²⁰⁵ Further, the scale of government investment required would likely make at least partial government ownership, and thus accountability, necessary (though the Max Planck Society operates in this independent-yet-government-funded manner).

2. EXISTING GOVERNMENT RESEARCH ORGANISATIONS

The lowest-friction approach could be to use existing UK government organisations, which could take one of two routes. The first would be to run Lovelace labs via UKRI. But its governance structure is not designed around the focused mission of fostering disruptive technoscience. It has a setup that might suit more conventional programmes, but can create barriers for those pursuing models that depart substantially from existing approaches; further, it does not have the important legal carve-outs that would be necessary for the Lovelace labs network, particularly around elements such as hiring rules and procurement freedoms.

The second option would be to expand ARIA, which does have the freedoms outlined above. This would be legally consistent with ARIA's existing primary legislation, but would likely overburden ARIA and compromise its autonomy. Lovelace labs and ARIA would share some attributes, but the former would require a distinct structure and mandate. Key differences include:

- **Infrastructure development.** Lovelace labs would move beyond distributing funds and grants, instead creating and nurturing long-term research communities based around physical labs and institutes.
- **Sustained commitment.** The network of Lovelace labs would need to secure stable funding commitments over timescales of 15 years and more (beyond the timescales of ARIA grants), attracting global talent by mitigating against funding uncertainty.

- **Complex infrastructure management.** Structures will need to be put in place to address the bureaucratic complexities associated with long-term physical infrastructure and community review.

3. A NEW, BESPOKE GOVERNMENT ORGANISATION

This involves creating what is, in effect, an ARIA for physical labs, which would likely create significantly more initial friction than the other two approaches, requiring primary legislation. However, it offers a combination of unique advantages, including:

- **The ability to adopt and extend ARIA-style freedoms.** Legislation could enable the institutional vehicle that sustains and coordinates Lovelace labs to operate outside conventional procurement and hiring rules (as with ARIA). It could also grant special freedoms relating to planning and the creation of bespoke audit and review processes, rather than inheriting standard government value-for-money processes.
- **A dedicated, bespoke governance arrangement tailored to purpose.** The coordinating institution could be staffed, structured and constituted – from its HR to its governing board – for the mission at hand, enabling specialisation and focus.
- **A strong signal to potential partners.** Establishing a new body signals strong government support, likely helping with subsequent partnerships and investment.

We favour this third option. The government is uniquely positioned to marry long-horizon planning with patient capital, and is the primary entity tasked with investing in the UK's long-term growth through science and technology. The new, bespoke government body could be called the Lovelace Society. This placeholder name and description as a society would emphasise the organisation's focus on empowering talent, pursuing technoscientific visions and cultivating research communities.

Recommendation: *The UK government should legislate for the creation of a new pioneer research entity (the Lovelace Society) dedicated to the creation and nurturing of a network of physical technoscientific research labs – and*

fund it sufficiently for 15 years of operations. By 2040 a small percentage of the public R&D budget, between 3 and 4 per cent, should be invested into the network of Lovelace labs.

The investment required is not significantly greater than that which supports projects such as the Catapult Network,²⁰⁶ which receives £320 million annually. As previously noted,²⁰⁷ securing this investment is not without its challenges – but they are challenges worth rising to in order to secure the future of frontier science and technology in the UK.

First Steps

In this section we outline the early steps to create an entity with the capabilities outlined above.

Our proposed path and objectives are structured in such a way that they degrade gracefully: even if the full Lovelace Society project proposed in this paper is not achieved, each of the three elements below is useful in its own right.

1. An initial research unit, otherwise known as the founding unit.
2. Funding that is optimised for the purpose required.
3. Primary legislation for the creation of an independent operating bureaucracy with properly aligned incentives.

We will examine these elements in turn. While they should be pursued in parallel, they are likely to mature on different timelines, roughly in the order presented.

1. THE FOUNDING UNIT

This unit is the brains of the early endeavour and its operational hub. It should operate similarly to a think-tank, taking the vision outlined in this report, adapting and extending it, then initiating steps to make it a reality.

The founding unit would be responsible for conducting studies to determine which research visions and nascent fields hold sufficient potential to merit the creation of a Lovelace lab. This would involve “field-strategist”²⁰⁸ scientists in the founding unit seeking individuals who could be candidate directors for subsequent labs. As part of this scoping exercise, it will produce vision documents: an early-stage process akin to a problem-finding phase, where the goal is to identify underexplored but promising lines of enquiry at the intersection of science and technology.

As well as providing potential founding visions for new laboratories, the resulting studies would be very useful to other institutions within the research ecosystem such as UKRI and ARIA and could lead to new startups. This kind of study, prominent in US national-security planning through the likes of JASON and RAND, is not yet prominent in UK R&D policy, and we are not aware of anything similar to the US-style field-strategist funding route in the UK.

The founding unit would also begin to assemble the Lovelace Society leadership team and further refine the operating model, leaving substantial latitude for evolution and for lab directors to adapt each lab’s setup as necessary. A CEO with executive decision-making authority would be required, who would be granted autonomy to select research areas independent of government intervention. Lessons from ARIA’s establishment²⁰⁹ underscore the importance of recruiting the initial Lovelace Society leadership team prior to legislation being passed, using a taskforce approach (in this case, that time-limited taskforce being the Lovelace Founding Unit) to avoid long delays. Full-time experts in science or technology, staffed externally to the government, should be at the centre of the programme. In parallel, the founding unit should begin convening a trustee board comprising people with experience of different R&D models.

Recommendation: *The government should announce a £3 million grant to produce a founding unit. This includes money to employ roughly eight field strategists for two years and a modest outlay on leadership, communications and an advisory board, as well as discretionary funds for workshops.*

2. FUNDING THAT IS OPTIMISED FOR THE PURPOSE REQUIRED

Establishing physical labs requires substantial investment. Current government investment in physical laboratories can often unintentionally hamstring those labs' leadership, both in terms of the amount spent and the conditions attached. In previous New National Purpose reports we have argued that this is a general trend across UK public R&D investment – and one that ultimately harms return on investment. For example, the Industrial Strategy Challenge Fund was criticised in 2021 by the National Audit Office for delays of more than a year in approving funding. In a competitive global race, such delays can mean that UK researchers and their collaborators get out of the blocks much later than their competitors.²¹⁰

Securing philanthropic support will depend, in part, upon the availability of a flexible pool of public funding that can be rapidly deployed to match or complement private contributions.

Here we outline five key requirements for an effective funding arrangement for an organisation such as the Lovelace Society, which we believe would require the creation of a dedicated, bespoke funding stream.

1. **Fifteen-year funding duration.** To enable Lovelace labs to attract a critical concentration of talent, carry out high-risk work and move beyond the setup phase with confidence, they need guaranteed, long-term support.
2. **Single business-case approval.** Rather than relying on government approval for each decision, a single business case should be approved outlining the mission the funding is tasked with achieving, with the organisation's board and subsequent external reviews overseeing more granular spend. This is how ARIA is set up.
3. **Sufficient funding for two or three major labs and some smaller endeavours.** About £250 million per year is needed to set up two or three labs funded to compete at the global frontier – comparable in scale and ambition to the Arc Institute – with a set of smaller, affiliate labs. This

estimate is based on an approximate annual budget of between £50 million and £80 million for a major, globally competitive laboratory, underpinned by an analysis of leading global institutions.²¹¹

4. **High level of discretion.** Labs should be able to decide how they allocate their funding. For example, they should be able to fund external teams without the need for central-government approval.²¹²
5. **Effective expert review, not audit.** Careful financial auditing is essential for good governance. However, effective oversight should not be conflated with requirements for extensive documentation for each internal decision regarding research, recruitment and investment. Rather, the performance of the laboratories should be based on a “review-not-audit” approach, with first-rate external international expertise evaluating each laboratory’s performance in accordance with the timescales specified below.

Recommendation: *The UK government should create a successor to the Research Ventures Catalyst at the 2026 Spending Review, this time with a view to the creation of lasting physical infrastructure. The precise details of this funding configuration would depend upon political appetite for the other steps outlined in this plan.*

3. PRIMARY LEGISLATION FOR CREATING A MISSION-FOCUSED, LEGALLY EMPOWERED TEAM

Establishing a new formal organisation to oversee the creation and management of new technoscientific laboratories is the optimal path to achieving the Lovelace vision.

This requires an act of Parliament, with a range of legal carve-outs closely modelled on the 2022 Advanced Research and Invention Agency Act.²¹³ The legislation would include exemptions from government procurement systems and a specific parliamentary remit for the mission of the society. In addition, oversight of the society’s investment decisions should be made by expert review, not via the conventional value-for-money framework, which requires primary legislation. This was the original intent for the ARIA bill.²¹⁴

This is why we have recommended that the UK government should pass primary legislation to create a Lovelace Society. It should be closely modelled on the 2022 Advanced Research and Invention Agency Act, and report into DSIT in a similar manner.

It is essential that the right people are recruited to leadership roles. The broad composition of the ARIA board is a useful starting point, but with greater emphasis on experience working in and running early-stage physical research institutions, avoiding the default failure mode of selecting grantees from within the existing ecosystem.

It should be straightforward to transfer legal and funding control of projects or laboratories from the existing infrastructure to the new Lovelace Society. Existing efforts – such as elements of ARIA programmes, standout university research groups or initiatives within other government agencies – could graduate into Lovelace physical research institutions where appropriate. In some cases it may be desirable to bring in established labs. Establishing clear, low-friction mechanisms for such transitions would enable the Lovelace project to grow organically, integrating promising teams and programmes into a unified ecosystem capable of driving breakthrough science at scale.

Recommendation: *The government should commission legal and policy work to investigate what legal carve-outs are necessary for projects or labs to transfer and integrate into the Lovelace Society.*

Potential Failure Modes and Evaluation

As Kay wrote to some of the authors of this paper, “As you know from looking at the past, most of the attempts to do something like ARPA-PARC ran afoul of a variety of human problems involving people trying to control instead of nurture, spreading the money around instead of concentrating on talent, trying to set problems rather than fund ‘problem finding’, and many more ways people have found to mess up good ideas and processes.”

We close by briefly considering first foreseeable pitfalls to avoid, and then the manner in which the Lovelace Society should be evaluated.

RESIST REVERSION TO THE STATUS QUO

Lovelace labs must resist being absorbed into the existing academic culture. The proposed proportion of R&D spending allocated to a potential network of Lovelace labs is very small in comparison to the amount allocated to academic institutions. The university department model would remain dominant, so there is a risk that Lovelace labs could be sucked back into the traditional culture – as seen at Janelia, where the gravitational pull of conventional academia has diluted the founding vision. If there is not sufficient scale or dynamism within the network of Lovelace labs to offer researchers a long-term career, with the likely outcome that they must return to academia, then the culture of academia will dominate in Lovelace.

Central to this autonomy is the deliberate empowerment of young, exceptional talent. The Lovelace network would succeed by capitalising on the disenfranchisement of early-career researchers within traditional academic structures. Although incorporating senior scientists from academia would undoubtedly enrich Lovelace, it would have to avoid becoming what would essentially be a university department without teaching, with the power brokers in the existing system migrating to the new system. According to this same principle, Lovelace labs would have to avoid prestige-based recruitment, whereby the most highly credentialed scientists or those with honours are appointed to run the organisation.

COUNTERACT BUREAUCRATIC DRIFT AND LOSS OF AGILITY

A defining risk is that the Lovelace Society could, over time, succumb to bureaucratic inertia, gradually losing the agency and dynamism central to its founding ethos. The growth of administrative layers and procedures, even when introduced with good intentions, can erode creativity and responsiveness.

As such, Lovelace labs would have to establish explicit early-warning mechanisms: regularly reviewing administrative overheads as a proportion of overall spending; conducting periodic internal audits of the “cost” of time that researchers spend fulfilling non-research tasks; and setting clear, predefined limits on administrative growth. Custodians should consciously keep decision-making close to the research frontline, rather than centralising power into hierarchical layers. Additionally, internal culture would have to remain sceptical of new administrative processes, demanding that every proposed bureaucratic measure justify itself against clear, measurable benefits to research outcomes. Lovelace’s periodic external reviews should explicitly assess administrative growth, to guard against creeping complexity and reaffirm the society’s fundamental commitment to streamlined, researcher-centric operations.

AVOID GENERIC OR STALE RESEARCH TOPICS

Another key consideration is the strategic selection of research topics. In identifying high-level topics for Lovelace labs, the society would have to avoid generic, amorphous research areas such as biotechnology, AI and quantum, which encourage generic hiring and dilute the potential for truly disruptive outcomes – at the same time bringing the labs into competition with established labs that are well funded via existing routes.

It is also essential that institutes not all be shoehorned into current national “priority areas”, limiting Lovelace labs to incremental progress rather than pioneering entirely new fields. AI was not a priority area when DeepMind was founded; nor was molecular biology when the LMB was created, nor personal computing when Xerox PARC came into existence. The aim of disruptive innovation is to create the next priority areas, not to exist within the current ones.

REJECT INSUFFICIENT SUSTAINED SUPPORT

A significant threat would arise from chronic underfunding, or “jam spreading”: trying to fund too many labs at insufficient funding levels. Underfunded labs would struggle to first attract the necessary talent, then to sufficiently empower that talent. Labs must be funded competitively against international comparators.

PREVENT STAKEHOLDER DILUTION

Effective stakeholder management is essential for preserving the purity of the Lovelace vision. Lovelace labs must not be beholden to a wide array of stakeholders, as overextension in such relationships can destroy focus and water down vision and ambition. The Lovelace Society’s primary stakeholders should be its researchers, along with the funders and taxpayers who support them.

Evaluation Framework

Given these potential failure modes, as well as the unusual freedoms we have advocated for the organisation, robust evaluation mechanisms are indispensable – and would have to be specifically designed to promote the society’s core aims. Clearly defined evaluation criteria would have to be established from the outset to preserve the unique character and ambitions of Lovelace.

Evaluation of the Lovelace Society would have to balance oversight with the freedom to innovate, and avoid the lure of the various metrics that dominate traditional research evaluation. Instead, evaluations should focus on alignment with long-term goals, organisational culture and transformative outcomes.

Our proposed framework comprises a balance of interim advisory reviews every five years with more comprehensive reviews every decade, both conducted by external, international expert groups that would visit the labs in person. These reviews would occur in addition to continual oversight by the society’s advisory and governance boards.

- **Every five years:** Light-touch, mid-cycle advisory reviews, available to the fellows, managers, directors and funders, will provide guidance for potential course correction.
- **Every ten years:** A comprehensive review addressing a set of core questions. It would be crucial to enshrine these evaluation criteria ahead of time to ensure that the usual standards of success are not unintentionally projected by external assessors.

Reviews should consider the following.

- Are Lovelace labs doing science fundamentally differently from the status quo, or have they reverted to competing based on traditional metrics and practices?
- Are they aligned with the core vision of empowering junior individuals and maintaining an egalitarian meritocracy?
- Are they global destinations for top-tier talent?
- Have Lovelace labs succeeded in nurturing a cohesive wider community, implementing ARPA-style internal promotion without succumbing to nepotism?
- Most importantly, have they produced valuable frontier research that would otherwise not have emerged?

06

Conclusion: A Call to Build the Future

The UK has long been a beacon of scientific discovery, home to breakthroughs that have reshaped the modern world. The Lovelace Society is a call to renew this spirit of possibility. It is not a proposal for reform – it is a proposal for creation.

By empowering the young, gathering and enabling our top scientific minds, and creating space for the next wave of visionary ideas, our hope is that Lovelace disruptive invention labs would spark a revitalisation of science in the UK. We seek to demonstrate that, with a little will, the country can recapture the energy of the Industrial Revolution and reorient itself towards Vannevar Bush’s “endless frontier”.

Progress thrives on diversity of approach. Without diversity, there can be no evolution. Only by embracing a variety of approaches can today’s challenges be met, ensuring that the UK’s scientific institutions keep up in a rapidly changing world.

The Lovelace project would be one experiment within this broader tapestry. We do not claim that our proposal is the best way to do science. What we can say with confidence is that there is no best way, and that overreliance on one single model is a major, perhaps *the* major challenge confronting science today. Ultimately, it will take a range of complementary approaches, enabling many kinds of scientific endeavour, to reimagine the pursuit of discovery and invention.

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Endnotes

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to be an accountant, you need to be a politician, you need to be a very skilled writer, an excellent communicator and all of those different things take up a bit of time, so that's possibly in second place why I will eventually leave the job. It's pulling you in too many directions ... That makes doing research tricky, because to really do the very best research, you need clear blocks of time. Having 40 minutes free, or an hour, just doesn't cut the mustard ... I haven't even mentioned management. As a university professor, you have to be a manager of people and projects."

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info@institute.global

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